Asymmetric Credit Access, Inflation, and Unemployment

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Abstract

We study how an empirically relevant asymmetry in credit access between employed and unemployed workers shapes the long-run relationship between inflation and unemployment. We add this asymmetry to a labour-market matching model with money in the framework of Berentsen, Menzio, and Wright (2011): employed workers settle a fraction of their goods-market trades on credit while unemployed workers transact in cash. Inflation acts through two channels of opposite sign: a Liquidity Channel that erodes firm profitability as cash trades shrink (raising unemployment), and an Employment Channel that raises the value of being employed relative to being unemployed as the cash-use asymmetry widens (lowering unemployment). At the calibrated parameters the two channels combine into a hump-shaped inflation-unemployment relationship: unemployment rises with inflation at low rates and falls at high rates. Three extensions—endogenous credit adoption, common credit access, and heterogeneous per-trade credit frictions—corroborate this mechanism.

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1. Introduction

We show that an empirically relevant asymmetry in goods-market credit access between employed and unemployed workers can, on its own, generate a non-monotone long-run relationship between inflation and unemployment. The asymmetry produces a hump-shaped curve in which unemployment first rises with inflation and then falls, without nominal rigidities and without the outside-option mechanism that earlier models of this kind have required. Our contribution is to identify this single structural ingredient as the source of the non-monotonicity, to decompose its effect into two opposing channels operating through inflation, and to show that the mechanism survives a sequence of natural perturbations of the underlying credit environment.

We incorporate this asymmetry into a labour-market matching model with money in the framework of Berentsen, Menzio, and Wright (2011). Employed workers settle a fraction of their goods-market trades on credit while unemployed workers transact only in cash; the rest of the model— bilateral bargaining over wages and goods-market prices, free entry of vacancies, money demand, matching—is standard. The only departure from Berentsen, Menzio, and Wright (2011) is the credit asymmetry; in particular, we do not endow unmatched workers with an outside option of any kind.

Liquidity Channel vs. Employment Channel. Inflation operates on equilibrium unemployment through two channels of opposite sign. The first, which we call the *Liquidity Channel*, is the standard force in this class of models: as inflation rises, the cost of holding cash for goods-market transactions rises, the quantity of cash trades shrinks, and firms expect lower profits from each cash trade. With lower expected profitability, firms post fewer vacancies and equilibrium unemployment rises with inflation. This channel pushes the long-run inflation–unemployment relationship upward and is present in any monetary-search labour model in which goods-market trades carry cash.

The second channel, which we call the *Employment Channel*, is new and is delivered by the credit asymmetry. Because employed workers can settle some trades on credit while unemployed workers cannot, the inflation-induced erosion of cash trades hits unemployed workers harder than employed ones. The relative value of being employed therefore rises with inflation, wages adjust downward in bilateral bargaining, and free entry brings more vacancies into the market. Equilibrium unemployment falls with inflation through this channel, opposing the Liquidity Channel head-on.

The two channels run in opposite directions, and their relative strength varies with the inflation rate. At low inflation the Liquidity Channel dominates and unemployment rises; at sufficiently high inflation the cumulative credit-access advantage of being employed builds enough that the Employment Channel takes over and unemployment falls. The crossover produces a hump-shaped inflation–unemployment relationship as the equilibrium outcome of the model, without recourse to nominal rigidities or to any exogenous wedge between the employed and unemployed states.

We then sharpen the mechanism through three extensions of the baseline. The first endogenises credit adoption on the employed side, letting the share of credit users respond to inflation through a one-time fixed adoption cost. The second relaxes the maintained assumption that unemployed workers transact only in cash and allows them to settle a fraction of trades on credit as well. The third introduces a per-trade variable cost on the credit margin that can differ between employed and unemployed buyers, capturing the empirical reality that unemployed borrowers typically face stricter terms and higher costs in formal credit markets.

The three extensions reshape the hump but do not eliminate the mechanism. Endogenous adoption pulls the peak forward and attenuates its height; allowing the unemployed to use credit monotonically flattens the curve and removes the hump only at the symmetric limit where the Employment Channel vanishes by construction; per-trade frictions on the unemployed credit margin, somewhat counter- intuitively, also flatten the hump rather than amplifying it, by making the unemployed value more sensitive than the employed value to credit-side distortions. The three exercises together show that the credit-access mechanism is robust to natural perturbations of the baseline assumptions and reorganises rather than overturns the hump.

The asymmetry the model rests on is empirically well documented. Herkenhoff (2019) shows, using credit-bureau records linked to unemployment-insurance microdata, that employed workers have substantially higher credit-card limits than unemployed workers and that consumer-credit access measurably lengthens unemployment durations. Sullivan (2008) documents from the Survey of Income and Program Participation that households accumulate unsecured debt during unemployment spells, with access rationed by prior credit history; and the Federal Reserve Bank of Atlanta's Diary of Consumer Payment Choice records that less-attached-to-the-labour-force households use cash for a substantially larger share of retail transactions than employed, higher-income households.[1] Despite its visibility, this asymmetry has not, to our knowledge, served as the basis for a theoretical account of the long-run inflation–unemployment relationship.

The implication is that the long-run inflation–unemployment relationship in this class of models is fundamentally a credit-access phenomenon. Once the buyer-side asymmetry is in place, the hump emerges from the interaction of inflation with credit structure rather than from nominal rigidities or expectational frictions, and the geometry of the hump traces out the dimensions of credit infrastructure that are likely to be empirically relevant for cross-country variation in the long-run inflation–unemployment trade-off.

[1] See Herkenhoff (2019), Sullivan (2008), and Greene, Schuh, and Stavins (2020) for methodological details on the credit-bureau, SIPP, and consumer-payment-survey data underlying each source.

Related literature. Our paper is part of the literature that asks whether long-run non-monotonicities in the inflation–unemployment relationship can be generated within the new-monetarist tradition. The closest precedent is Berentsen, Menzio, and Wright (2011), which embeds Kiyotaki–Wright bilateral trade inside a Mortensen–Pissarides labour market and shows that a hump-shaped relationship emerges when unmatched workers have a positive outside option that interacts with the cash-distortion term. Subsequent work has revisited the Berentsen, Menzio, and Wright (2011) mechanism along several directions. Bethune and Rocheteau (2023) introduce an interest-rate channel through which inflation reduces the supply of public liquid assets and lowers the required real return on assets, raising the present value of firm profits and encouraging job creation; Bethune et al. (2024) reintroduce a hump by giving consumers an outside option to search for another seller in the frictional goods market; Ait Lahcen et al. (2022) extend Berentsen, Menzio, and Wright (2011) to a stochastic environment to account for the nonlinear unemployment effect of the inflation tax; Lehmann (2021) explores a closely related cash-in-advance formulation; and Nosal and Rocheteau (2017) provides a textbook treatment of the money-and-labour-market integration that underlies the entire programme. Our contribution relative to this strand is to identify a single structural ingredient—an empirically documented asymmetry in goods-market credit access between employed and unemployed workers—that generates the same non-monotonicity without an outside option, and to trace it to a clean two-channel decomposition that subsequent extensions of the credit structure can be read against.

The non-monotonicity question itself has been studied from several angles, and we relate our wage-bargaining force to two precedents in particular. Liu (2008) proposes an early mechanism that is closely related in spirit to our Employment Channel: the unemployed have a higher matching probability in the goods market and therefore carry more cash than the employed, so inflation hurts them more and wages adjust downward in bargaining. We differ in three respects. First, we maintain the Berentsen, Menzio, and Wright (2011) timing convention and the Kiyotaki–Wright bilateral structure rather than competitive matching, which lets us derive bargaining and quantity terms in closed form. Second, the source of differential cash use in our model is access to credit rather than matching probabilities, which maps directly onto consumer-finance microdata on credit-card limits by employment status. Third, we work out the comparative statics along three credit-environment dimensions (endogenous adoption, common access, heterogeneous frictions) that have no counterpart in Liu (2008). The most closely related contemporaneous work is Gu, Wang, and Jiang (2025), who develop a new-monetarist model with Mortensen–Pissarides labour and identify a two-channel structure— a wage-bargaining channel and a cash-financing channel—that closely parallels our Liquidity and Employment Channels. They use a competitive goods market and focus on the time-series slope variation in the post-2000 U.S. data; we use a bilateral goods market in the Berentsen, Menzio, and Wright (2011) timing convention and focus on the hump shape of the relationship and the cross-sectional credit asymmetry that drives it. Rocheteau, Rupert, and Wright (2007) and Dong (2011) show that the sign of the inflation–unemployment relationship can depend on the substitutability of centralized- and decentralized-market goods in the utility function, while Gomis-Porqueras, Huangfu, and Sun (2020) document a hump-shaped relationship between inflation and the capital stock rather than unemployment.

A separate strand integrates labour-market search with frictional credit. Bethune, Rocheteau, and Rupert (2015) extend the Mortensen–Pissarides model with limited-commitment consumer credit and derive endogenous borrowing limits that interact with labour-market outcomes; they show that the long-run expansion of U.S. unsecured consumer credit between 1978 and 2008 accounts for roughly seventy percent of the secular decline in long-term unemployment. We share the integrated labour-and-credit-market framework but differ in two respects: we maintain differential credit access as a structural primitive rather than deriving credit limits from limited commitment, and our central object is the inflation–unemployment relationship rather than the unemployment–credit joint dynamics. On the firm side of credit, Wasmer and Weil (2004), Petrosky-Nadeau and Wasmer (2013), and Petrosky-Nadeau (2013) study how frictions in the financing of vacancy posting affect the cyclical volatility of labour markets. Herkenhoff (2019) documents at the household level the link between consumer-credit access and unemployment durations that motivates the asymmetric primitive we maintain here.

Our quantitative results speak to a long-standing empirical literature documenting variation in the slope of the long-run Phillips curve across countries and time periods. King and Watson (1994) report slope variation in U.S. data across decades; Berentsen, Menzio, and Wright (2011) find a positive low-frequency relationship in HP-filtered U.S. data and Haug and King (2014) confirm this finding using band-pass filters. In contrast, Karanassou, Sala, and Snower (2010) document a negative inflation–unemployment relationship in a panel of European countries, and Dolado and Jimeno (1997) document a sign switch in Spanish data. The mechanism we develop generates either sign of the slope locally depending on which channel dominates at the prevailing inflation rate, and accommodates this empirical heterogeneity by tracing it to cross-country differences in the underlying credit infrastructure.

The remainder of the paper is organised as follows. Section 2 sets up the model. Section 3 derives the two-channel decomposition and presents the numerical baseline. Section 4 works through the three extensions. Section 5 concludes. Appendix A collects the proofs; Appendix B records the calibration.

2. Model

This section develops the model. The environment is the standard Berentsen, Menzio, and Wright (2011) economy with one financial asymmetry: employed workers settle a fraction of their decentralised goods-market trades on credit, while unemployed workers do not. We study how this asymmetry shapes the long-run relationship between inflation and unemployment. The exposition proceeds in the order in which the model is solved: the primitives (Section 2.1) and the sequence of events (Section 2.2); the goods- and labor-market bargaining problems and the free-entry condition (Section 2.3); the household and firm value functions (Sections 2.4–2.5); and the stationary monetary equilibrium (Section 2.6), which collects the equilibrium conditions and states the uniqueness result on which the rest of the paper relies.

2.1. Primitives

Populations. Time is discrete and indexed by $t = 0, 1, \ldots$, with all agents discounting at common rate $\beta \in (0, 1)$. The economy is populated by a unit measure of *workers*; let $u_t \in [0, 1]$ denote the mass of unemployed at the start of period t, so that $1 - u_t$ are employed. Workers are ex ante identical, risk neutral, and infinitely lived. *Firms* are atomistic and identical; entry is free, so the mass of vacancies v_t is determined endogenously by the zero-profit condition derived in Section 2.3.

Goods and unit of account. Two goods circulate: a perishable specialised KW good with quantity q, and a non-storable AD good that serves as the numéraire for wages, dividends, and credit redemption. Money is fiat and perfectly divisible.

Functional forms. The KW utility $u(q)$ and production cost $c(q)$ are continuously differentiable on $(0, \infty)$, with $u' > 0$, $u'' < 0$, $c' \geq 0$, $c'' \geq 0$, and standard Inada-type endpoint conditions $u'(0^+) = +\infty$, $c'(0^+) \in [0, 1]$. These ensure that the efficient KW quantity $q^\star$ defined by $u'(q^\star) = c'(q^\star)$ exists and is unique. The parametric forms used in calibration are specified in Appendix B.

Labor matching. Contacts in the labor market are governed by a constant-returns matching function $M(u, v)$, strictly increasing and concave in each argument, over the mass u of searching unemployed and the mass v of posted vacancies. With tightness $\theta \equiv v/u$, the job-finding and vacancy-filling rates depend only on θ,

$$\lambda_h(\theta) \equiv \frac{M(u, v)}{u} = M(1, \theta), \qquad \lambda_f(\theta) \equiv \frac{M(u, v)}{v} = \frac{\lambda_h(\theta)}{\theta},$$

truncated at one if necessary, with λ_h increasing and λ_f decreasing in θ. An employed worker separates exogenously with probability $\delta \in (0, 1)$.

Goods-market meetings and credit. In the KW goods market a buyer meets a seller with exogenous probability α_h and a seller meets a buyer with probability α_0, both constants in $(0, 1)$ (Liu 2008); Berentsen, Menzio, and Wright (2011) instead derive them from a KW matching function so that they depend on the unemployment rate, whereas treating them as constants keeps the goods-market block decoupled from labor-market quantities. Meetings differ in the settlement technology available, following Bethune et al. (2024). In a *monetary meeting* there is no monitoring technology, so trade is quid pro quo and the buyer settles in cash (money); in a *credit meeting* a record-keeping technology supports enforced repayment in the following AD, so the buyer may settle on credit. Credit availability is state-dependent on the buyer's post-MP employment status—the model's central new ingredient. An employed buyer is in a monetary meeting with probability $\omega_e \in (0, 1)$ and in a credit meeting with probability $1 - \omega_e$; an unemployed buyer is always in a monetary meeting, $\omega_u = 1$, so she has no credit access in the baseline. Section 4 relaxes both restrictions in turn.

2.2. Sequence of events

Within-period timing. Each period is partitioned into three sub-periods, ordered

$$\underbrace{\text{MP}}_{\text{Mortensen–Pissarides matching}} \longrightarrow \underbrace{\text{KW}}_{\text{Kiyotaki–Wright goods trade}} \longrightarrow \underbrace{\text{AD}}_{\text{Arrow–Debreu centralized market}}.$$

This is the timing convention introduced by BMW. The three sub-periods serve the following purposes.

(MP) The *Mortensen–Pissarides labor matching* sub-period. Firms have committed vacancies v at the close of the previous AD; unemployed workers (mass u) search and match through the technology $M(u, v)$ defined above. An unemployed worker finds a job with probability $\lambda_h(\theta)$; an employed worker loses hers with probability δ.[2] The post-MP employment state $j' \in \{0, 1\}$ is the one observed by counterparties in the ensuing KW meeting.

(KW) A continuum of anonymous bilateral meetings opens between workers (buyers) and firms (sellers), with the matching probabilities α_h, α_0 and the monetary/credit meeting split described above. Conditional on a match, the firm produces a perishable specialized good at quantity q and the two parties bargain over (q, d), where d is the

[2] In stationary equilibrium these transition rates imply the Beveridge identity $\delta(1 - u) = \lambda_h(\theta)u$, stated formally as equation (42) in Section 2.6.4; free entry in Section 2.3 then determines θ and closes the labor block.

buyer's payment—settled in cash in a monetary meeting, or as a credit claim on the next AD in a credit meeting.

(AD) The centralized Arrow–Debreu market. Three blocks of activity occur. *(i) Wages and benefits:* matched workers receive the bargained wage w (Section 2.6); unemployed workers receive the flow value b. *(ii) Monetary policy:* the nominal money supply grows at the constant gross rate $1+\pi$, with new money distributed lump-sum to all workers; in equilibrium the nominal interest rate i on a one-period bond satisfies the Fisher relation $1+i=(1+\pi)/\beta$. *(iii) Asset rebalancing and free entry:* agents settle any credit incurred in KW, consume the numéraire, and choose their real-balance carryover $m' \geq 0$ for the next MP sub-period; firms simultaneously post vacancies at flow cost k until expected profit drives entry to zero.

2.3. Bargaining and free entry

2.3.1. Goods market: KW bargaining

Two distinct bargaining problems arise: the bilateral KW match sets both the quantity q and the buyer's payment d, while the labor match in AD sets the wage w. We use *Kalai proportional* bargaining throughout, with seller share $\eta \in (0,1)$ in the KW match and worker share $\gamma \in (0,1)$ in the wage. In the KW match Kalai delivers the tractable linear payment schedule $g(q) = \eta u(q) + (1-\eta)c(q)$ derived below and avoids the buyer-surplus non-monotonicity that generalized Nash bargaining generates once the liquidity constraint binds. For the wage, where the AD budget constraint (13) makes the bargaining frontier linear in the numéraire, Kalai and generalized Nash with the same share coincide.

KW cash bargaining. In a cash meeting at quantity q with payment d, the buyer's surplus is $u(q)-d$ and the seller's is $d-c(q)$. Kalai with seller share η requires

$$u(q)-d = (1-\eta)\big[u(q)-c(q)\big], \tag{1}$$

$$d-c(q) = \eta\big[u(q)-c(q)\big], \tag{2}$$

which together pin down the cash payment schedule

$$g(q) \equiv \eta\, u(q) + (1-\eta)\, c(q). \tag{3}$$

With $i>0$, the buyer never wishes to carry slack cash, so the liquidity constraint binds in equilibrium and the realised payment equals her real-balance position: $d=m=g(q)$. The buyer's and seller's per-trade surpluses are therefore

$$\mathcal{S}^{mb}(q) = u(q)-g(q) = (1-\eta)\big[u(q)-c(q)\big], \qquad \mathcal{S}^{mf}(q) = g(q)-c(q) = \eta\big[u(q)-c(q)\big]. \tag{4}$$

KW credit bargaining. In a credit meeting there is no liquidity constraint and (in the baseline) no credit cost. Kalai bargaining selects the efficient quantity $q^\star$ solving $u'(q^\star) = c'(q^\star)$ and splits total surplus proportionally:

$$\mathcal{S}^{cb} = (1-\eta)\big[u(q^\star) - c(q^\star)\big], \qquad \mathcal{S}^{cf} = \eta\big[u(q^\star) - c(q^\star)\big]. \tag{5}$$

2.3.2. Labor market: wage bargaining

Wage bargaining. The bargained wage w maximises the Kalai/Nash objective with worker share γ. The first-order condition is

$$\gamma J^f = (1-\gamma)\big(\mathcal{U}_1 - \mathcal{U}_0\big), \tag{6}$$

where J^f is the firm's filled-match value (10) and $\mathcal{U}_1 - \mathcal{U}_0$ the worker's lifetime employment premium, both constructed in the equilibrium of Section 2.6; the wage is solved there once these surpluses are in hand.

2.3.3. Free entry

Asset equations. Let J^v denote the value of an open vacancy and J^f the value of a filled match, both evaluated at the start of a period (pre-MP). A vacancy posted at the previous AD is matched in the current MP with probability $\lambda_f(\theta)$, and a filled match dissolves at separation rate δ in the next MP. The Bellman equations in steady state are

$$J^v = -k + \beta\big[\lambda_f(\theta)\, J^f + (1-\lambda_f(\theta))\, J^v\big], \tag{7}$$

$$J^f = \big(y - w\big) + R(i,u,\theta) + \beta\big[(1-\delta)\, J^f + \delta\, J^v\big], \tag{8}$$

where $R(i,u,\theta)$ is the firm's expected per-period KW profit and w the bargained wage, both determined in the equilibrium of Section 2.6.

Free entry. Imposing $J^v = 0$ in equilibrium, the system (7)–(8) collapses to

$$k = \beta\,\lambda_f(\theta)\, J^f, \tag{9}$$

$$J^f = \frac{y - w + R(i,u,\theta)}{A}, \qquad A \equiv 1 - \beta(1-\delta). \tag{10}$$

Combining (9)–(10) gives the firm-side free-entry condition

$$k = \frac{\beta\,\lambda_f(\theta)}{A}\big[\, y - w + R(i,u,\theta)\,\big]. \tag{11}$$

The wedge $1/A$ is the standard DMP discount that capitalises the flow surplus $(y - w + R)$ of a filled match net of separation.

2.4. Households

State and value functions. A worker's relevant state at the start of period t is the pair (j, m_+), where $j \in \{0, 1\}$ is the employment state inherited from period $t-1$ (0 = unemployed, 1 = employed) and $m_+ \geq 0$ is the real balance carried into period t's MP — selected as the next-period balance in the prior AD sub-period. Throughout, we write m_+ for the AD choice of next-period balance and re-use the same symbol when that balance enters MP one sub-period later. Following the BMW convention, we index value functions by the sub-period at which they are evaluated: $\mathcal{U}_j(m_+)$ denotes her value at the start of MP with state j and balance m_+; $\mathcal{V}_{j'}(m_+)$ the value at the start of KW with post-MP state j' and the same balance; and $\mathcal{W}_{j'}(m)$ the value at the start of AD with state j' and post-KW balance m (after any cash payment made in KW). All three are determined recursively as we step backward through the three sub-periods of the period.

AD optimization. A worker entering AD with post-KW balance m and state j' chooses consumption x of the numéraire and the real-balance $m_+ \geq 0$ to carry into next period's MP. Her AD Bellman equation is

$$\mathcal{W}_{j'}(m) = \max_{x,\, m_+ \geq 0} \Big\{ x + (1 - j')\,\ell + \beta\,\mathcal{U}_{j'}(m_+) \Big\} \tag{12}$$

subject to the period-t AD budget constraint

$$x + \rho\, m_+ = m + y_{j'} + \Pi - T, \qquad y_1 = w,\ y_0 = b, \tag{13}$$

where $\ell \geq 0$ is the flow utility from home production or leisure when unemployed, Π is the dividend rebate from firms (Section 2.5), T a lump-sum tax that finances unemployment benefits and absorbs the monetary transfer (Section 2.1), and $\rho \equiv 1 + \pi = \beta(1 + i)$ (by the Fisher relation) is the current-period real cost of carrying one unit of next-period real balance — equivalently, the gross inflation factor. Substituting x from (13) into (12) gives the reduced AD problem

$$\mathcal{W}_{j'}(m) = m + I_{j'} + \max_{m_+ \geq 0} \big\{ -\rho\, m_+ + \beta\,\mathcal{U}_{j'}(m_+) \big\}, \qquad I_{j'} \equiv y_{j'} + (1 - j')\,\ell + \Pi - T. \tag{14}$$

Linearity of $\mathcal{W}_{j'}$. The maximand in (14) does not involve m, so

$$\mathcal{W}_{j'}(m) = m + \mathcal{W}_{j'}(0), \qquad \mathcal{W}'_{j'}(m) = 1, \tag{15}$$

a linearity result that mirrors Lagos and Wright (2005) and BMW (2011, equation 2). The result requires only quasi-linearity in the numéraire, which holds because x enters $\mathcal{W}_{j'}$ linearly in (12). It has two consequences that drive everything that follows. *First,* the AD optimisation can be solved separately from the rest of the period: each state-j' chooses $m^*_{+,j'} \equiv \arg\max_{m_+}\{-\rho m_+ + \beta\mathcal{U}_{j'}(m_+)\}$ once and for all in steady state. *Second,* because the maximand depends on $\mathcal{U}_{j'}$ but not on $\mathcal{U}_j$ for $j \neq j'$, employed and unemployed AD-choosers in general carry different real balances. We denote these choices by $m_{+,1}$ and $m_{+,0}$ and refer to j in $m_{+,j}$ as the worker's *money type*. In BMW's symmetric-credit baseline the choices coincide, $m_{+,0} = m_{+,1}$; the asymmetric-credit restriction $\omega_u = 1$ of Section 2.1 is what generates the wedge $m_{+,1} \neq m_{+,0}$ and is the model's primary channel of state dependence.

KW value. A worker entering KW with post-MP state j' and balance m_+ meets a seller with probability α_h. Conditional on a match, she trades in cash with probability $\omega_{j'}$ at quantity q^m (paying $g(q^m)$) and in credit with the complementary probability $1-\omega_{j'}$ at the efficient quantity $q^\star$. Let $\mathcal{S}^{mb}(q) \equiv u(q) - g(q)$ denote the buyer's cash surplus and $\mathcal{S}^{cb}$ her credit surplus; the explicit Kalai forms of $g(\cdot)$ and $\mathcal{S}^{cb}$ are derived in Section 2.3. The KW Bellman equation is

$$\begin{aligned}\mathcal{V}_{j'}(m_+) \;=\; &\alpha_h\,\omega_{j'}\big[u(q^m) + \mathcal{W}_{j'}(m_+ - g(q^m))\big] \;+\; \alpha_h(1-\omega_{j'})\big[\mathcal{S}^{cb} + \mathcal{W}_{j'}(m_+)\big]\\ &+ (1-\alpha_h)\,\mathcal{W}_{j'}(m_+).\end{aligned} \tag{16}$$

Substituting the linearity result (15) into (16) collapses to

$$\mathcal{V}_{j'}(m_+) \;=\; m_+ + \mathcal{W}_{j'}(0) \;+\; \alpha_h\big[\omega_{j'}\,\mathcal{S}^{mb}(q^m) + (1-\omega_{j'})\,\mathcal{S}^{cb}\big], \tag{17}$$

where the term m_+ is the liquidation value of carried balances and the bracketed term is the expected KW gain.

Credit and the AD debt state. A buyer who settles in a credit meeting consumes $u(q^\star)$ now and carries her payment as a one-period debt $d = g(q^\star)$ into the following AD, where the monitoring technology enforces repayment (Section 2.3). Carrying this obligation as an explicit AD state, the budget constraint (13) gains a term $-d$, so by the linearity (15) the debt enters additively,

$$\mathcal{W}_{j'}(m, d) \;=\; \mathcal{W}_{j'}(m, 0) - d, \tag{18}$$

and the optimal balance $m^*_{+,j'}$ is independent of d. The credit branch of (16) is then $u(q^\star) + \mathcal{W}_{j'}(m_+, d) = \mathcal{S}^{cb} + \mathcal{W}_{j'}(m_+)$, with $\mathcal{S}^{cb} = u(q^\star) - d$: the debt washes out of the state and is absorbed into the credit surplus. We therefore carry only (j, m_+) as the worker's state; the credit position never enters as a separate state variable. This is the standard within-period treatment of enforced credit under quasi-linear preferences.

MP transition. The MP sub-period draws the labor shock: the post-MP state j' has distribution

$$\Pr[j' = 1 \mid j = 1] \;=\; 1 - \delta, \qquad \Pr[j' = 1 \mid j = 0] \;=\; \lambda_h(\theta), \tag{19}$$

with complementary probabilities placing weight on $j' = 0$. Averaging over the post-MP state,

$$\mathcal{U}_j(m_+) \;=\; \sum_{j' \in \{0,1\}} \Pr[j' \mid j]\, \mathcal{V}_{j'}(m_+). \tag{20}$$

KW gain and employment premium. The baseline restriction $\omega_u = 1$ implies that the credit branch of (17) is active only when $j' = 1$. Substituting (17) into (20) and evaluating at the optimal money type $m_+ = m_{+,j}$ for each entering state j, the lifetime entering-MP value decomposes as

$$\mathcal{U}_1(m_{+,1}) = m_{+,1} + (1 - \delta)\mathcal{W}_1(0) + \delta\, \mathcal{W}_0(0) + \Omega_1(i, \theta), \tag{21}$$

$$\mathcal{U}_0(m_{+,0}) = m_{+,0} + \lambda_h(\theta)\mathcal{W}_1(0) + (1 - \lambda_h(\theta))\mathcal{W}_0(0) + \Omega_0(i, \theta), \tag{22}$$

where the entering-state-j *expected KW gain* is

$$\Omega_1(i, \theta) = \alpha_h \big\{[\delta + (1 - \delta)\omega_e]\, \mathcal{S}^{mb}(q_1^m) + (1 - \delta)(1 - \omega_e)\, \mathcal{S}^{cb}\big\}, \tag{23}$$

$$\Omega_0(i, \theta) = \alpha_h \big\{[\lambda_h(\theta)\omega_e + 1 - \lambda_h(\theta)]\, \mathcal{S}^{mb}(q_0^m) + \lambda_h(\theta)(1 - \omega_e)\, \mathcal{S}^{cb}\big\}, \tag{24}$$

with $q_j^m \equiv q^m(m_{+,j})$. The KW component of the worker's employment premium is

$$\Delta(i, \theta) \;\equiv\; \Omega_1(i, \theta) - \Omega_0(i, \theta). \tag{25}$$

Δ is positive in the asymmetric-credit baseline ($\omega_u = 1$) because the employed buyer ($j = 1$) retains credit access in a fraction $1 - \omega_e$ of her KW meetings while the unemployed buyer ($j = 0$) does not. Δ enters wage bargaining in Section 2.6 through the worker's outside option and is the source of the *employment-premium channel* (Channel B) of the long-run inflation–unemployment relationship developed in Section 3.1.

AD first-order condition. Differentiating (14) with respect to m_+ at the interior optimum yields a state-specific Euler equation for each entering type:

$$\rho = \beta\, \mathcal{U}_1'(m_{+,1}), \tag{26a}$$

$$\rho = \beta\, \mathcal{U}_0'(m_{+,0}). \tag{26b}$$

The state-dependence of $\mathcal{U}_j$ across $j \in \{0, 1\}$ is what gives rise to two distinct money-demand equations in Section 2.6.

Marginal value of money. Differentiating (17) with respect to m_+ at the buyer's binding cash constraint $g(q^m) = m_+$ (hence $dq^m/dm_+ = 1/g'(q^m)$) and noting that the credit branch contributes nothing to $\mathcal{V}_{j'}'(m_+)$ because $\mathcal{S}^{cb}$ is evaluated at the efficient quantity $q^\star$, independent of the buyer's real balance,

$$\mathcal{V}_{j'}'(m_+) \;=\; 1 \;+\; \alpha_h\, \omega_{j'} \left[\frac{u'(q^m)}{g'(q^m)} - 1 \right]. \tag{27}$$

This is the envelope-theorem expression for the marginal value of money at the post-MP state j'. Aggregating across MP transitions via (20),

$$\mathcal{U}_j'(m_+) \;=\; 1 \;+\; \alpha_h\, w_j \left[\frac{u'(q^m)}{g'(q^m)} - 1 \right], \tag{28}$$

where the entering-state-j *expected cash-use weight* is

$$w_1 \;\equiv\; \delta + (1-\delta)\,\omega_e, \qquad w_0 \;\equiv\; \lambda_h(\theta)\,\omega_e + 1 - \lambda_h(\theta). \tag{29}$$

Each weight is the probability that the carried balance $m_{+,j}$ is actually spent on cash next period: an AD-employed worker remains employed with probability $1-\delta$ (using cash with probability ω_e) or separates with probability δ (cash-only, since $\omega_u = 1$), giving w_1; an AD-unemployed worker finds a job with probability $\lambda_h(\theta)$ (cash with probability ω_e) or remains unemployed (cash-only), giving w_0.

Equation (28) is the envelope condition that the AD FOC (26) feeds on. Combined with the Fisher relation $\rho/\beta = 1+i$ (from $\rho \equiv \beta(1+i)$), it yields the two money-demand equations MD-1 and MD-0 derived in Section 2.6.

2.5. Firms

Firms are risk-neutral, discount the future at β, and atomistic. Each filled match produces output y in the AD sub-period; each open vacancy carries a flow cost $k \geq 0$ until matched. Free entry forces the value of posting a vacancy to zero in equilibrium.

KW seller profit. A filled-match firm acts as a seller in the KW sub-period: with probability α_0 it meets a buyer, whose identity is the joint draw of her pre-MP state $j \in \{0,1\}$ (equivalently her "money type", since her real balance was chosen one AD ago when her state was j, and pins the cash quantity q_j^m) and her post-MP state $j' \in \{0,1\}$ (her current employment status, which keys her cash/credit choice). In steady state, the joint mass on the four KW cells is

$$\begin{aligned} \mu_{1,1} &= (1-u)(1-\delta), \quad & \mu_{1,0} &= (1-u)\,\delta, \\ \mu_{0,1} &= u\,\lambda_h(\theta), & \mu_{0,0} &= u\big[1-\lambda_h(\theta)\big], \end{aligned} \tag{30}$$

with the Beveridge identity (42) implying $\mu_{1,1} + \mu_{1,0} = 1-u$ and $\mu_{0,1} + \mu_{0,0} = u$. Letting $\mathcal{S}^{mf}(q)$ denote the seller's bargaining surplus in a cash trade at quantity q and $\mathcal{S}^{cf}$ her surplus in a credit trade at $q^\star$ (Section 2.3), the per-filled-job expected KW profit per period is

$$R(i,u,\theta) \;=\; \alpha_0 \sum_{(j,j')\in\{0,1\}^2} \mu_{j,j'} \left\{\omega_{j'}\, \mathcal{S}^{mf}(q_j^m) + (1-\omega_{j'})\, \mathcal{S}^{cf}\right\}. \tag{31}$$

Substituting $\omega_u = 1$ and collapsing the credit terms via the identity $(1-u)(1-\delta) + u\lambda_h(\theta) = 1-u$ yields the compact form used throughout the analysis,

$$\begin{aligned} R(i,u,\theta) \;=\; \alpha_0 \Big\{ & (1-u)\big[\delta + (1-\delta)\omega_e\big]\, \mathcal{S}^{mf}(q_1^m) \\ & + u\big[\lambda_h(\theta)\,\omega_e + 1 - \lambda_h(\theta)\big]\, \mathcal{S}^{mf}(q_0^m) \\ & + (1-u)(1-\omega_e)\, \mathcal{S}^{cf} \Big\}. \end{aligned} \tag{32}$$

The state-1 and state-0 weights in (32) have the same form as in the worker's KW gain Ω_j of equations (23)–(24), reflecting the same underlying joint distribution over (j_m, j'). The credit-trade coefficient collapses to $(1-u)(1-\omega_e)$ because employed-buyer credit trades are the only credit transactions in the baseline; the credit mass is therefore independent of where these employed buyers *end up* after MP, so long as the Beveridge identity holds.

From firm-side to joint surplus. The wage w that closes (11) is determined by Kalai bargaining in Section 2.6, jointly with the worker's value functions of Section 2.4. The bargained wage satisfies $y - w + R = (1-\gamma)\,A\,S/D$, where $\gamma \in (0,1)$ is the worker's bargaining share, $D \equiv (1-\gamma)A + \gamma B(\theta)$ with $B(\theta) \equiv A + \beta\,\lambda_h(\theta)$, and the joint match surplus is

$$S(i,u,\theta) \;\equiv\; y - b + R(i,u,\theta) + \Delta(i,\theta). \tag{33}$$

Substituting into (11) yields the compact free-entry condition used in the equilibrium analysis,

$$k = \Xi(\theta)\,\lambda_f(\theta)\,S(i,u,\theta), \qquad \Xi(\theta) \equiv \frac{\beta\,(1-\gamma)}{(1-\gamma)A + \gamma\,B(\theta)}. \tag{34}$$

Economically, (34) says that firms post vacancies until the cost k equals their expected discounted share $\Xi(\theta)\lambda_f(\theta)$ of the joint surplus S. Monetary policy reaches the labor market exclusively through $S(i,u,\theta)$: a higher nominal interest rate i contracts cash trades and lowers R, while widening the gap between cash and credit surpluses raises Δ. These two forces push S — and therefore the equilibrium θ — in opposite directions.

Two features of (34) drive the equilibrium structure of Section 2.6. *First*, the wage w has been eliminated, so θ is determined by (i,u) alone given (32). *Second*, combined with the Beveridge identity (42), u is in turn a function of θ, collapsing the equilibrium system to a single equation in θ — which we solve by bisection, exploiting the θ-independence of MD-1 (36).

2.6. Stationary Monetary Equilibrium

We now assemble the stationary monetary equilibrium from the bargaining rules of Section 2.3 and the value functions of Sections 2.4–2.5, in three steps: (i) the AD first-order condition (26) together with the KW payment schedule (3) yields the two money-demand equations MD-1 and MD-0; (ii) the wage bargaining rule (6) is solved for the equilibrium wage; (iii) the free-entry condition (11) is recast in compact form. The result is summarised in Definition 1.

2.6.1. Money demand

The household's AD first-order condition (26) and the envelope expression (28) of Section 2.4 together imply

$$\rho = \beta\,\mathcal{U}'_j(m_{+,j}) = \beta\left\{1 + \alpha_h\,w_j\big[u'(q_j^m)/g'(q_j^m) - 1\big]\right\}, \qquad j \in \{0,1\}, \tag{35}$$

where the state-dependent cash-use weights w_1, w_0 are given by (29) and the cash quantity satisfies the binding liquidity constraint $m_{+,j} = g(q_j^m)$. Substituting $\rho = \beta(1+i)$ from (13) into (35) and dividing by β gives $1 + i = 1 + \alpha_h w_j[u'(q_j^m)/g'(q_j^m) - 1]$, which rearranges to

$$i = \alpha_h\left[\delta + (1-\delta)\omega_e\right]\left[\frac{u'(q_1^m)}{g'(q_1^m)} - 1\right] \qquad \text{(MD-1)} \tag{36}$$

$$i = \alpha_h\left[\lambda_h(\theta)\,\omega_e + 1 - \lambda_h(\theta)\right]\left[\frac{u'(q_0^m)}{g'(q_0^m)} - 1\right] \qquad \text{(MD-0)} \tag{37}$$

the two money-demand equations of the model.

The two equations differ only through w_j, the probability that the carried balance is spent on cash in next-period KW. An AD-employed worker ($j = 1$) is likely to stay employed and access credit, so $w_1 = \delta + (1-\delta)\omega_e$ is small; an AD-unemployed worker ($j = 0$) is likely to remain unemployed and use cash, so $w_0 = \lambda_h(\theta)\omega_e + 1 - \lambda_h(\theta)$ is large. The resulting wedge — $m_{+,1} < m_{+,0}$ at the calibrated parameters — is the asymmetric-credit mechanism at the heart of the model.

2.6.2. Wage and labor market

Worker employment premium. Using the MP-transition identity (20), the linearity of $\mathcal{W}_{j'}$ in m, and the definition of Ω_j in Section 2.4 (equations (23)–(24)), the worker's lifetime employment premium satisfies the steady-state recursion

$$\mathcal{U}_1 - \mathcal{U}_0 \;=\; (w-b) + \Delta(i,\theta) + \beta\big[(1-\delta) - \lambda_h(\theta)\big]\big(\mathcal{U}_1 - \mathcal{U}_0\big), \tag{38}$$

where $w - b$ is the AD income gap and $\Delta = \Omega_1 - \Omega_0$ the KW employment premium (25). The latter captures the employed worker's additional KW gain from her credit access: an AD-employed worker expects to retain credit access (probability $1 - \omega_e$) while an AD-unemployed worker is essentially cash-only. Solving,

$$\mathcal{U}_1 - \mathcal{U}_0 \;=\; \frac{w - b + \Delta(i,\theta)}{B(\theta)}, \qquad B(\theta) \;\equiv\; 1 - \beta(1-\delta) + \beta\,\lambda_h(\theta). \tag{39}$$

A direct derivation of (38) from the KW Bellman equations of Section 2.4 (in particular the simplification $\mathcal{V}_{j'}(m_+) = m_+ + \mathcal{W}_{j'}(0) + \alpha_h \,\text{KW gain}_{j'}$) and the envelope condition at the worker's optimal real balance is given in Appendix A.

Wage formula. Substituting $J^f = (y - w + R)/A$ from (10) and $\mathcal{U}_1 - \mathcal{U}_0 = (w - b + \Delta)/B(\theta)$ from (39) into the bargaining FOC (6) and solving for w,

$$w(i,u,\theta) \;=\; \frac{\gamma\,B(\theta)\big[y + R(i,u,\theta)\big] + (1-\gamma)\,A\big[b - \Delta(i,\theta)\big]}{(1-\gamma)\,A + \gamma\,B(\theta)}. \tag{40}$$

2.6.3. Compact free-entry condition

The wage (40) allows us to eliminate w from the firm-side free-entry condition (11). Substituting (40) into $y - w + R$ and simplifying,

$$y \;-\; w(i,u,\theta) \;+\; R(i,u,\theta) \;=\; \frac{(1-\gamma)\,A}{(1-\gamma)\,A + \gamma\,B(\theta)}\;S(i,u,\theta), \tag{41}$$

where $S(i, u, \theta) = y - b + R + \Delta$ is the joint match surplus (33). Inserting (41) into $k = \beta\lambda_f(\theta)(y - w + R)/A$ yields the compact free-entry condition already stated in Section 2.5,

$$k = \Xi(\theta)\,\lambda_f(\theta)\,S(i, u, \theta), \qquad \Xi(\theta) \equiv \frac{\beta\,(1-\gamma)}{(1-\gamma)\,A + \gamma\,B(\theta)}.$$

The wage (40) is bracketed by two limits. When $\gamma \to 0$ (zero worker bargaining power), $w \to b - \Delta(i, \theta)$ and the firm captures the entire match surplus; when $\gamma \to 1$ (full worker power), $w \to y + R(i, u, \theta)$ and the worker captures it. Inflation enters (40) directly through $R(i, u, \theta)$ (the firm-side liquidity channel of Section 3.1) and $\Delta(i, \theta)$ (the employment-premium channel); u and θ further respond to i in equilibrium, amplifying both effects.

2.6.4. Definition of stationary monetary equilibrium

Before stating the definition, we record the steady-state flow balance in the labor market. With separations at rate δ for the employed and job-finding at rate $\lambda_h(\theta)$ for the unemployed, the unemployment rate is stationary when inflows equal outflows:

$$\delta\,(1-u) = \lambda_h(\theta)\,u. \tag{42}$$

We refer to (42) as the Beveridge identity. Together with the compact free-entry condition (34), it closes the labor block and pins down (u, θ) for given (R, Δ, w).

DEFINITION 1. *For a given nominal interest rate $i \geq 0$, a* stationary monetary equilibrium *(SME) is a tuple $(q_1^m,\ q_0^m,\ \Delta,\ R,\ w,\ u,\ \theta)$ such that*

(i) q_1^m *solves MD-1, equation* (36)*;*
(ii) q_0^m *solves MD-0, equation* (37)*;*
(iii) $\Delta(i, \theta)$ *is given by* (25) *with* Ω_j *from* (23)–(24)*;*
(iv) $R(i, u, \theta)$ *is given by* (32)*;*
(v) $w(i, u, \theta)$ *is given by* (40)*;*
(vi) (u, θ) *jointly solve the Beveridge identity* $\delta(1-u) = \lambda_h(\theta)u$ *in* (42) *and the free-entry condition* (34).

Definition 1 leaves two questions open: whether such a tuple exists, and whether it is unique. The next proposition answers both — existence for every admissible nominal interest rate $i \in [0, \bar{i})$ and uniqueness under a mild monotonicity condition on the free-entry residual.

PROPOSITION 1 (Existence and uniqueness of stationary monetary equilibrium). *Define the upper bound on the nominal interest rate at which money is valued,*

$$\bar{i} \equiv \alpha_h\,\omega_e\,\frac{1-\eta}{\eta}.$$

Fix $i \in [0, \bar{i})$ and let the maintained assumptions on $(u, c, \lambda_h, \lambda_f)$ of Section 2.1 and on g of Section 2.3.1 hold. Let the free-entry residual be

$$\mathcal{F}(\theta; i) \equiv k - \Xi(\theta)\lambda_f(\theta)\,\mathcal{S}\big(i, u(\theta), \theta\big), \qquad u(\theta) \equiv \frac{\delta}{\delta + \lambda_h(\theta)}.$$

(a) *An SME exists.*

(b) *The SME is unique whenever the elasticity bound*

$$\frac{1}{S}\frac{\partial S(i, u(\theta), \theta)}{\partial \theta} < \frac{1-\sigma}{\theta} + \frac{\sigma\gamma\beta\lambda_h(\theta)}{\theta\,[A + \gamma\beta\lambda_h(\theta)]} \tag{43}$$

holds at every $\theta \in (0, \infty)$.

The proof is in Appendix A.1.

COROLLARY 1 (BMW limit). *At $\omega_e = 1$ the SME is unique for every $i \in [0, \bar{i})$.*

PROOF. At $\omega_e = 1$, the cash-use weights satisfy $w_1 = w_0(\theta) = 1$, so MD-1 and MD-0 coincide and $q_1^m = q_0^m = q^m(i)$, independent of θ. Hence $\Delta \equiv 0$ and $R(i, u, \theta) = \alpha_0\,\mathcal{S}^{mf}(q^m(i))$ is independent of (u, θ). Therefore $S(i, u(\theta), \theta) = y - b + R(i)$ is θ-independent and $\partial S/\partial\theta = 0$, which satisfies the elasticity bound of Proposition 1(b). □

COROLLARY 2 (Friedman rule). *At $i = 0$ the SME is unique.*

PROOF. At $i = 0$ both MD-1 and MD-0 give $h(q_j^m) = 0$, so $q_1^m = q_0^m = q^\star$. Evaluated at $q^\star$, the Kalai surpluses collapse to $\mathcal{S}^{mb}(q^\star) = \mathcal{S}^{cb}$ and $\mathcal{S}^{mf}(q^\star) = \mathcal{S}^{cf}$, so cash and credit trades yield identical buyer and seller surpluses. Substituting into (23)–(24) gives $\Omega_1 = \Omega_0 = \alpha_h\mathcal{S}^{cb}$ and hence $\Delta = 0$. Substituting into (32), the Beveridge identity $(1-u)\delta = u\lambda_h(\theta)$ collapses the bracketed weight to unity, yielding $R = \alpha_0\,\mathcal{S}^{cf}$, independent of (u, θ). Therefore $\partial S/\partial\theta = 0$, satisfying the elasticity bound of Proposition 1(b). □

3. The Non-Monotone Inflation-Unemployment Relationship

This section establishes the paper's main result: inflation has a non-monotone effect on long-run unemployment. Section 3.1 characterises analytically the two opposing channels through which inflation reaches the labor market — a liquidity channel that raises unemployment and an employment-premium channel that lowers it — and shows how their relative strength shifts with the level of inflation. Section 3.2 verifies the mechanism at the calibrated baseline and reports the headline quantitative results (peak inflation, hump magnitude, and welfare cost).

3.1. Liquidity Channel vs. Employment Channel

Inflation operates through two competing channels. The *Liquidity Channel* is common to Berentsen, Menzio, and Wright (2011): higher inflation compresses KW cash quantities and lowers the firm's goods-market profit R, reducing vacancy creation and raising unemployment. The *Employment Channel* is new: because employed workers settle a fraction $1 - \omega_e$ of KW transactions on credit, the AD-employed type anticipates lower cash use than the AD-unemployed type. The resulting asymmetry in real balances produces an employment premium Δ that *changes sign* as inflation rises, generating the hump. When $\omega_e \to 1$, credit access shuts down, $\Delta \equiv 0$, and only the Liquidity Channel remains, recovering the monotone BMW limit (Corollary 3 below). Section 3.2 verifies the mechanism quantitatively.

Liquidity Channel. The Liquidity Channel operates by raising the opportunity cost of cash, shrinking KW cash trades, and lowering the firm's KW profit R. Both AD-employed and AD-unemployed households hold cash for next-period KW; higher inflation raises the nominal interest rate i, increasing the opportunity cost of carrying money. From the money-demand equations (36)–(37), both cash-trade quantities fall strictly with i: implicit differentiation gives $\partial q_j^m/\partial i < 0$ on the interior monetary regime, since $h(q) = u'(q)/g'(q) - 1$ is strictly decreasing on $(0, q^\star]$ (Step 1 of Appendix A.1). The firm's KW profit $R(i, u, \theta)$ in (32) sums two cash-trade contributions, evaluated at q_1^m and q_0^m, and a credit-trade contribution $(1 - u)(1 - \omega_e)\mathcal{S}^{cf}$ that is independent of i. As cash quantities shrink with inflation, only the cash-trade contributions fall, and the firm's profit per filled job declines.

LEMMA 1 (Liquidity Channel). *For any* $(u, \theta) \in (0, 1) \times (0, \infty)$ *and* $i \in (0, \bar{i})$, $\partial R(i, u, \theta)/\partial i < 0$.

The proof is in Appendix A. Along the equilibrium path, $R^\star(\pi)$ is verified strictly decreasing in π over the calibrated baseline; see Figure 1, left panel. The decline in R compresses the joint match surplus, reduces vacancy creation through free entry (34), and raises equilibrium unemployment. This channel is present in both BMW and the current model.

Employment Channel. The Employment Channel arises because, under the BMW timing, the AD-employed and AD-unemployed types face different expected cash-use frequencies in next-period KW. The two money-demand equations (36)–(37) differ in their cash-use weights $w_1 = \delta + (1-\delta)\omega_e$ and $w_0(\theta) = \lambda_h(\theta)\omega_e + 1 - \lambda_h(\theta)$ defined in (29): the same nominal interest rate i prices a different effective cost of cash for the two types. The weights differ because of the timing MP $\to$ KW $\to$ AD: an AD-employed household, having chosen $m_{+,1}$ this period, retains employment in next-period KW with probability $1 - \delta$ (using cash with probability ω_e) or separates with probability δ (using cash with certainty); an AD-unemployed household finds a job with probability $\lambda_h(\theta)$ (using cash with probability ω_e) or stays unemployed (using cash with certainty). Whenever the equilibrium satisfies the standard active-labor-market condition $u > \delta$—equivalently $\lambda_h(\theta) < 1 - \delta$—the AD-employed household has a lower expected cash-use frequency than the AD-unemployed one, so $w_1 < w_0(\theta)$.

LEMMA 2 (Quantity Ranking). *Suppose the equilibrium satisfies* $u > \delta$*, equivalently* $\lambda_h(\theta) < 1 - \delta$*, and* $\omega_e \in (0, 1)$*. Then for any* $i \in (0, \bar{i})$*,* $q_1^m(i, \theta) < q_0^m(i, \theta) \leq q^\star$*, with equality if and only if* $i = 0$ *or* $\omega_e \to 1$.

The proof is in Appendix A. Lemma 2 says that the AD-employed type carries *less* cash than the AD-unemployed type in the calibrated region, recovering the conventional intuition that employed workers, who expect to use credit more often, economise on cash holdings.

The employment premium decomposes into a cash-trade surplus gap (negative) and a credit-access advantage (positive):

$$\Delta(i, \theta) = \alpha_h\Big[w_1 \mathcal{S}^{mb}(q_1^m) - w_0(\theta) \mathcal{S}^{mb}(q_0^m) + \big((1 - \delta) - \lambda_h(\theta)\big)(1 - \omega_e) \mathcal{S}^{cb}\Big], \tag{44}$$

which follows from (25), (23), and (24). The first two terms form the cash-trade surplus differential between the AD-employed and AD-unemployed types, and the third is the credit-access advantage of the AD-employed type, who is more likely to use credit in next-period KW when $\lambda_h(\theta) < 1 - \delta$. The behaviour of Δ as inflation varies is characterised informally as follows.

Three regimes of $\Delta(i, \theta)$. *At the Friedman rule* ($i = 0$), the cash-trade and credit-access components cancel exactly, giving $\Delta(0, \theta) = 0$. Indeed $q_1^m = q_0^m = q^\star$ and $\mathcal{S}^{mb}(q^\star) = \mathcal{S}^{cb}$; substituting into (44) and using the identity $w_1 - w_0(\theta) = (1 - \omega_e)[\lambda_h(\theta) - (1 - \delta)]$ from Lemma 2 produces the cancellation.

For small $i > 0$, the AD-employed type's cash-surplus deficit exceeds the AD-unemployed type's, so $\Delta(i, \theta) < 0$ at leading order i^2. The mechanism is asymmetric: by Lemma 2, $q_1^m < q_0^m$ implies $q^\star - q_1^m > q^\star - q_0^m$, so the AD-employed type loses more cash-trade surplus. Since the credit-access term in (44) does not depend on i, this asymmetric deficit breaks the Friedman-rule cancellation. A second-order Taylor expansion around $i = 0$ under $\lambda_h(\theta) < 1 - \delta$ makes this precise: $\Delta(i, \theta)$ is strictly negative for all sufficiently small $i > 0$.

For large i, cash trades collapse and only the credit-access advantage survives, pushing $\Delta > 0$. Specifically, q_j^m approach zero at the upper end of the interior monetary regime, so the cash surpluses $\mathcal{S}^{mb}(q_j^m)$ vanish; the credit term in (44) dominates, leaving $\Delta \approx \alpha_h(1 - \omega_e)[(1 - \delta) - \lambda_h(\theta)]\mathcal{S}^{cb} > 0$ under $\lambda_h(\theta) < 1 - \delta$. Continuity of Δ in i then forces at least one sign crossing at some $i^\dagger \in (0, \bar{i})$: the inflation tax shifts from hurting the

AD-employed type at low i to favouring her at high i. Whether this crossing is unique, and whether Δ is monotone in i beyond it, are properties of the parameter region; in the calibrated baseline of Section 3.2 both hold.

The entire sign pattern of Δ is conditional on the active-labor-market condition $u > \delta$ (equivalently $\lambda_h(\theta) < 1 - \delta$) along the equilibrium path. This condition is satisfied at the calibrated baseline; outside it—for instance when $u < \delta$—the sign of Δ may reverse.

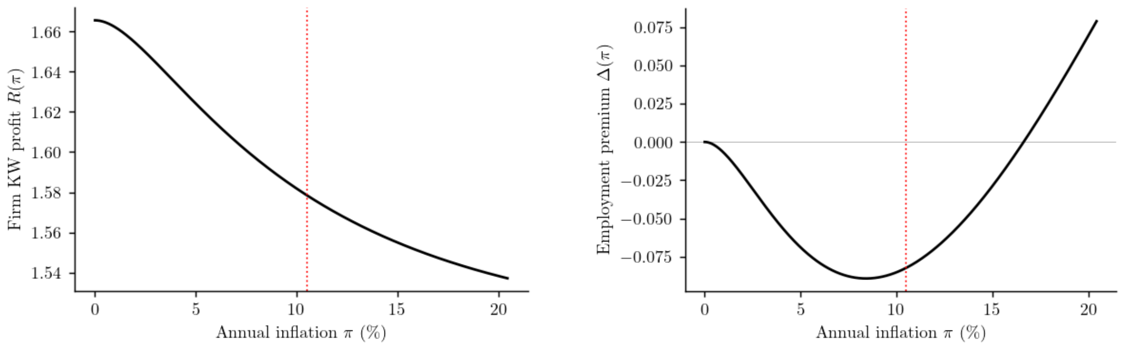
Liquidity vs. Employment. Having characterised each channel separately, we now compare them. The Liquidity Channel pushes unemployment monotonically upward as R declines with i. The Employment Channel switches sign at $i^\dagger$: below $i^\dagger$ it reinforces the Liquidity Channel (both push u up), while above $i^\dagger$ it offsets the Liquidity Channel (working against the rise in u). The shape of the inflation–unemployment relationship is therefore determined by the relative strength of the two channels along the equilibrium path.

A higher employment premium lowers the bargained wage, providing the transmission from Δ to unemployment. By the Kalai wage formula (40), $\partial w/\partial\Delta = -(1-\gamma)A/D < 0$: $\Delta \uparrow \Rightarrow w \downarrow$ widens the match surplus $y - w + R$, posting becomes more profitable, and unemployment falls. Combined with the directional analysis above, $\Delta < 0$ at low i amplifies the Liquidity Channel's rise in $u^\star$, while $\Delta > 0$ at high i moderates and eventually reverses it.

The sign reversal transmits to the shape of the inflation–unemployment curve through the joint match surplus $S(i, u, \theta) = y - b + R + \Delta$: by free entry (34), higher S induces more vacancy posting, tighter labor markets, and lower unemployment. By the implicit function theorem applied to the free-entry condition, the sign of $du^\star/di$ is the opposite of the sign of $\partial S/\partial i$. Near the Friedman rule, both $\partial R/\partial i < 0$ and $\partial\Delta/\partial i < 0$, so $\partial S/\partial i < 0$ and unemployment rises with inflation. As i increases, Δ reaches an interior minimum and $\partial\Delta/\partial i$ turns positive: the Employment Channel begins to offset the Liquidity Channel even while Δ remains negative. The curve peaks at the interior rate $i^\star$ where $\partial\Delta/\partial i\big|_{i=i^\star} = -\partial R/\partial i\big|_{i=i^\star}$; beyond $i^\star$, the Employment Channel dominates and unemployment falls. The zero-crossing $i^\dagger$, where $\Delta = 0$, is a separate landmark that may lie above or below $i^\star$: what drives the hump is the rate at which the employment premium rises, not the sign of its level. Section 3.2 documents this quantitatively.

COROLLARY 3 (BMW Limit). *As $\omega_e \to 1$: $\Delta(i, \theta) \to 0$ for all i, $S \to y - b + R$, and $u^\star(\pi)$ converges to the monotone BMW limit.*

The proof is in Appendix A. The corollary confirms that asymmetric credit access—the single departure from BMW—is the sole source of the hump. When $\omega_e = 1$, employed workers face the same cash environment as unemployed workers: both money-demand weights collapse to $w_1 = w_0 = 1$, Lemma 2 gives $q_1^m = q_0^m$, the Employment Channel shuts



A. Liquidity Channel: firm KW profit $R(\pi)$, declining monotonically.

B. Employment Channel: employment premium $\Delta(\pi)$, starting at zero, dipping negative, and crossing zero at $\pi^{\dagger} \approx 17\%$.

FIGURE 1. Channel decomposition at the calibrated baseline. Dotted vertical line: $\pi^{\star} \approx 10\%$ (unemployment peak).

down entirely, and the inflation–unemployment curve reverts to the monotone BMW relationship.

3.2. Numerical baseline

We calibrate the model to the U.S. economy at quarterly frequency with annual reporting. Eight free parameters $(\eta, \alpha, A_u, b, k, Z, \omega_e, \gamma)$ are jointly chosen to match eight data moments: the goods-market markup, the slope of money demand $\partial \log(M/Y)/\partial i$, the long-run money-to-output ratio, the average unemployment rate at the Friedman rule, the unemployment-to-wage ratio b/w, labor-market tightness, the share of employed buyers paying cash, and the vacancy-posting cost. Appendix B reports the targets, the calibrated parameter values, the moment-matching residuals, and the numerical procedure.

At the calibrated baseline, the long-run inflation–unemployment relationship is hump-shaped with peak unemployment $u^{\star}(\pi^{\star}) = 6.77\%$ at annual inflation $\pi^{\star} = 10.42\%$. The rise from the Friedman-rule level $u_{FR} = 6.27\%$ is $\Delta u = 0.506$ percentage points (Figure 2). The peak lies inside the moderate-inflation range routinely studied in the monetary-policy literature and well below the upper bound $\bar{i}$ of the interior monetary regime in Proposition 1. As $\omega_e \to 1$, the curve collapses to the monotone BMW relationship (Corollary 3); the calibrated $\omega_e = 0.044$ is therefore the empirically relevant departure from BMW that produces the hump.

The two-channel decomposition of Section 3.1 is verified quantitatively in Figure 1. The left panel confirms that $R^{\star}(\pi)$ is strictly decreasing in π (Lemma 1, Liquidity Channel). The right panel confirms that $\Delta^{\star}(\pi)$ is non-monotone: it vanishes at $\pi = 0$, dips negative on $(0, \pi^{\dagger})$ with $\pi^{\dagger} \approx 17\%$, and turns positive thereafter (Employment Channel). The peak inflation rate $\pi^{\star} \approx 10\%$ sits *below* the zero-crossing $\pi^{\dagger}$: consistent with the discussion of

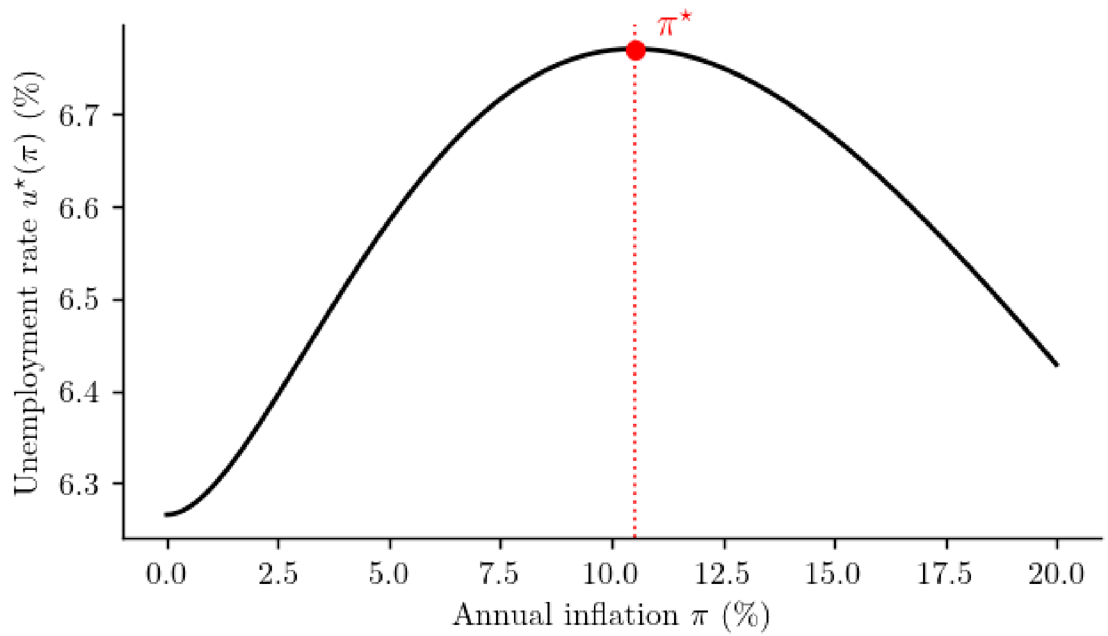


FIGURE 2. Calibrated long-run inflation–unemployment relationship $u^\star(\pi)$ at the 8×8 baseline. Peak at $\pi^\star = 10.42\%$ with rise $\Delta u = 0.506$ pp from the Friedman-rule level $u_{FR} = 6.27\%$. Dotted vertical line: $\pi^\star$.

Section 3.1, the hump turns where the *rate* of improvement in Δ overtakes the continuing decline in R, not where Δ changes sign.

Appendix C reports the welfare cost of inflation at the baseline and in the BMW limit.

4. Extensions

The baseline of Section 3.2 treats three primitives as fixed: the employed worker's cash probability ω_e, the unemployed worker's cash probability $\omega_u = 1$, and the absence of credit cost. Each extension relaxes one of these and examines how the hump responds. Section 4.1 endogenises ω_e through a one-time adoption cost, so that the share of credit users responds to inflation; the resulting curve shifts left and flattens. Section 4.2 allows ω_u to take any value in $(0, 1]$, generating common credit access across employment status; the hump shrinks monotonically as ω_u falls and disappears entirely at the symmetric limit $\omega_u = \omega_e$. Section 4.3 introduces per-trade credit costs ϕ_j that distort the credit quantity by buyer type, and asks how three focal scenarios (symmetric, employed-only, and unemployed-only frictions) reshape the hump. The three subsections together map the depth, breadth, and quality of credit infrastructure.

4.1. Endogenous credit adoption

Endogenising ω_e as a function of inflation strengthens the Liquidity Channel: when cash is more costly to hold, more employed workers pay the fixed cost to adopt credit, and the share of cash users $\omega_e(i)$ falls with i.

All other primitives are identical to the baseline of Section 3.2. Each employed worker draws a personal one-time adoption cost $\kappa_i \sim \text{Exp}(1/\kappa_{\max})$ and adopts credit if and only if the per-period gain from credit use exceeds her draw. The gain is the expected surplus from substituting away from cash trade in the employed-state KW,

$$G(i) \;\equiv\; \alpha_h\,(1-\delta)\big[\mathcal{S}^{cb} - \mathcal{S}^{mb}(q_1^{m,\text{ref}}(i))\big], \tag{45}$$

where $q_1^{m,\text{ref}}(i)$ denotes the cash quantity evaluated at the baseline reference $\omega_e = \omega_e^{\text{ref}}$. The adoption rate is $f(G) = 1 - \exp(-G/\kappa_{\max})$, and the aggregate cash probability is the complement,

$$\omega_e(i) \;=\; \exp\!\big(-G(i)/\kappa_{\max}\big). \tag{46}$$

Higher inflation lowers ω_e through a single cash-trade margin. As i rises, the employed worker's reference cash quantity $q_1^{m,\text{ref}}$ shrinks, the cash-trade surplus $\mathcal{S}^{mb}$ falls, and the gain from holding credit G rises; more workers therefore pay the adoption cost, and the share of cash users $\omega_e(i)$ declines. Evaluating G at the reference quantity rather than at the equilibrium q_1^m shuts down the positive feedback through ω_e and guarantees a unique equilibrium for every i; the richer fixed-point version is left for separate analysis.[3]

Endogenous adoption brings the peak forward and flattens the hump (Figure 3). The peak appears at lower inflation, with $\pi^\star$ shifting from 10.5% at the baseline to 7.0%; the amplitude shrinks from $\Delta u = 0.51$ to 0.41 percentage points; and the curve runs strictly below the baseline at every inflation rate above the new peak.

The driving force is an inflation-induced substitution toward credit. As inflation rises, holding cash becomes costlier, more employed workers pay the adoption cost, and the equilibrium cash share $\omega_e(\pi)$ falls smoothly from one at the Friedman rule (Figure 4). This substitution reinforces both channels of Section 3.1 in the same direction: the Liquidity Channel weakens because cash trades shrink as a share of KW activity, so the firm KW profit R declines more gradually with π; the Employment Channel strengthens because the credit-access term in (44) scales with $(1-\omega_e(\pi))$, which rises with inflation. Both forces accelerate the inflation rate at which the gain in Δ overtakes the loss in R—hence the earlier peak—and both push the curve below the baseline at high inflation through the

[3]The single free parameter $\kappa_{\max}$ is pinned down by requiring $\omega_e(\bar{\imath}) = \omega_e^{\text{ref}} = 0.044$ at the calibration nominal rate (annual inflation 5.82%). The exponential form gives a closed-form $\kappa_{\max} = G(\bar{\imath})/(-\ln \omega_e^{\text{ref}}) = 0.495$ and Ext 1 nests the baseline of Section 3.2 exactly at $\pi = \bar{\pi}$.

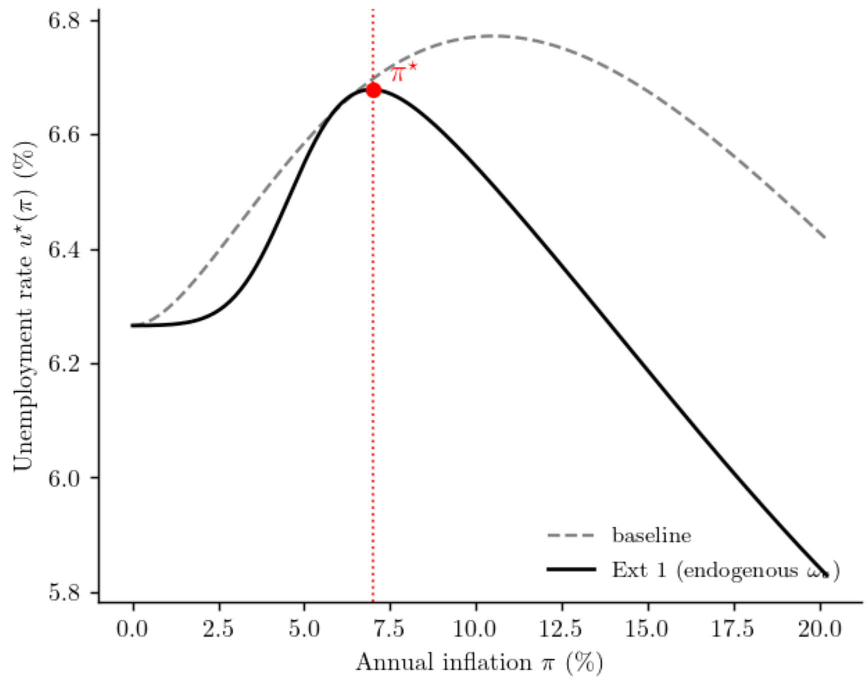


FIGURE 3. Long-run inflation–unemployment curve under Ext 1 (solid) versus the fixed-ω_e baseline (dashed).

rising credit-access advantage of being employed, which lowers wages and raises vacancy creation.

4.2. Common credit access

This subsection turns to common credit access: the unemployed worker's cash-use probability ω_u is freed from its baseline value of 1 and allowed to take any value in $[\omega_e, 1]$, so both employment types share access to credit, only by different amounts. The exercise asks what happens to the long-run inflation–unemployment relationship as the asymmetry between the two types closes.

The cash-use weight w_j in (29) is the probability that a worker leaving AD in employment state $j \in \{0, 1\}$ trades with cash rather than credit in the next KW ($j = 1$ for employed, $j = 0$ for unemployed). With a free ω_u, these weights generalize to

$$w_1 = \delta\,\omega_u + (1-\delta)\,\omega_e, \qquad w_0(\theta) = \lambda_h(\theta)\,\omega_e + \big(1-\lambda_h(\theta)\big)\,\omega_u, \tag{47}$$

which reduce to the baseline at $\omega_u = 1$ and coincide ($w_1 = w_0$) at $\omega_u = \omega_e$. All other equations of Section 2.6 carry over unchanged. The model nests the baseline at $\omega_u = 1$ exactly, and at $\omega_u = \omega_e$ the employment premium $\Delta(i, \theta)$ vanishes identically.

The hump shrinks monotonically as ω_u falls and disappears at the symmetric limit

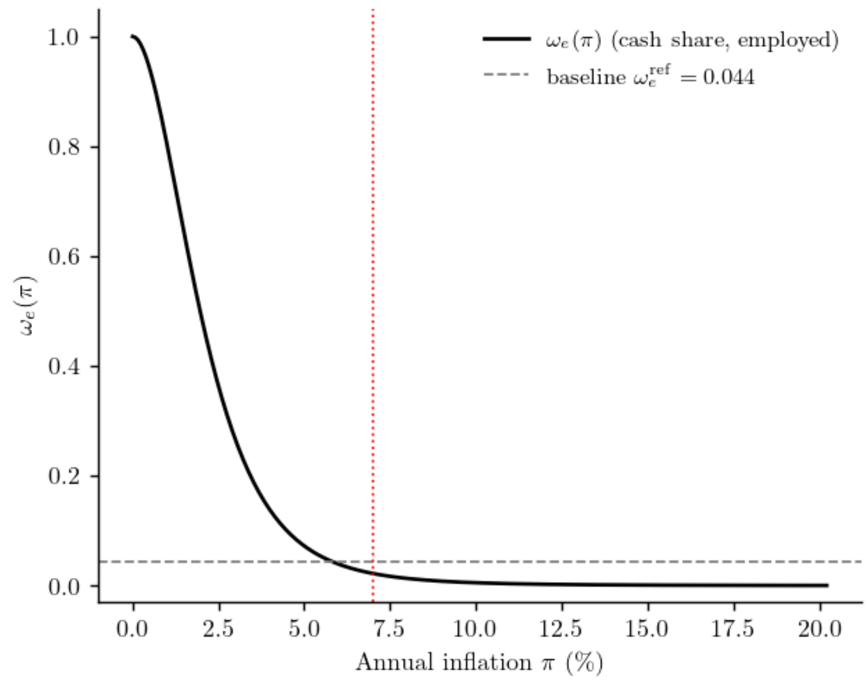


FIGURE 4. Equilibrium cash probability $\omega_e(\pi)$ under Ext 1.

(Figure 5). Amplitude moves from $\Delta u = 0.51$ percentage points at the baseline through a flattest non-trivial value of 0.12 pp at $\omega_u \approx 0.10$, and the peak migrates leftward in tandem, from $\pi^\star = 10.5\%$ at the baseline to $\pi^\star \approx 6\%$ at the flattest hump. Below $\omega_u \approx 0.07$ the interior peak slides off the right edge of the inflation range, and at $\omega_u = \omega_e$ the curve is monotone increasing on the entire range.

The shrinkage is the gradual death of the Employment Channel. As ω_u falls, the cash-use asymmetry $w_1 - w_0$ closes; Δ flattens in inflation and the credit-access advantage of being employed loses its bite. Once $\omega_u = \omega_e$, $\Delta \equiv 0$ and only the Liquidity Channel remains: a monotone fall in $R(\pi)$ shrinks aggregate surplus, vacancy creation contracts, and $u^\star(\pi)$ rises smoothly with inflation, with no offsetting force to bend the curve back down. The hump is a strict creature of the Employment Channel, and equalizing credit access kills it by killing that channel.

4.3. Heterogeneous credit frictions

Sections 4.1 and 4.2 vary *who* uses credit—the depth of credit on the employed side (Ext 1) and the breadth of credit across employment types (Ext 2). This subsection turns to the *quality* of credit: holding the population's credit-use shares fixed, what changes if the credit transaction itself is costly, and what changes if those costs differ by who is borrowing? To pose the question precisely we start from a partial-equalization environment in which

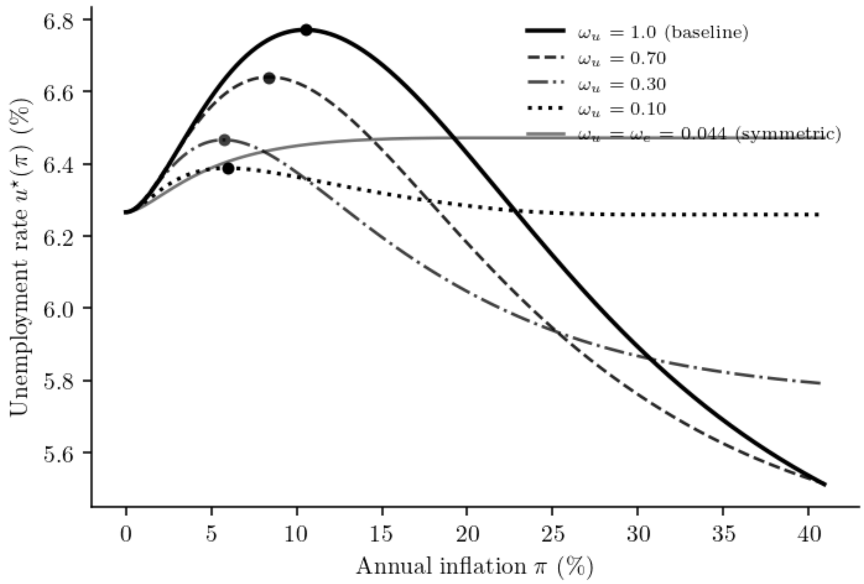


FIGURE 5. Long-run inflation–unemployment relationship under varying ω_u. Filled dots mark interior peaks (true humps); the symmetric-limit curve has no interior peak and is monotone over the inflation range.

both employment types use credit to some degree—concretely $\omega_u = 0.5$, the midpoint of the range studied in Section 4.2. In this environment half of the unemployed and almost all of the employed transact on credit, so a per-trade credit cost has bite on both sides of the market. We then add a marginal cost on each credit meeting, borne by the buyer, and ask which side of the cost matters more for the long-run inflation–unemployment relationship.

Three focal scenarios isolate the buyer-side asymmetry: a *symmetric* case in which all credit trades carry the same cost; an *employed-only* case in which costs fall solely on the employed side; and an *unemployed-only* case in which costs fall solely on the unemployed side. The last scenario is the empirically natural one—unemployed and low-income borrowers typically face higher interest rates and stricter terms in credit-market data—and it produces a striking counter-intuitive result: concentrating credit costs on the unemployed side *shrinks* the long-run macro hump rather than amplifying it. The reversal traces to a weight imbalance between the two sides of the credit margin, which we make precise next.

Each credit meeting between a type-j buyer ($j \in \{e, u\}$) and a seller incurs a marginal cost ϕ_j per unit traded, borne by the buyer. The Kalai bargaining problem maximises a modified joint surplus

$$T_j(q) = u(q) - c(q) - \phi_j q, \tag{48}$$

giving the first-order condition

$$u'(q_c^j) \;=\; c'(q_c^j) + \phi_j, \tag{49}$$

which collapses to $q_c^j = q^\star$ at $\phi_j = 0$ and shrinks the credit quantity below $q^\star$ for any $\phi_j > 0$. The seller-share split $(\eta, 1-\eta)$ is unchanged, so the credit-margin surpluses become $\mathcal{S}_j^{cb} = (1-\eta)T_j(q_c^j)$ and $\mathcal{S}_j^{cf} = \eta\, T_j(q_c^j)$. Because the credit terms in the employment premium Δ and the firm KW profit R now split by buyer type, the four credit weights are

$$\begin{aligned} \omega_e^{c,1} &= (1-\delta)(1-\omega_e), \quad & \omega_u^{c,1} &= \delta(1-\omega_u), \\ \omega_e^{c,0} &= \lambda_h(\theta)(1-\omega_e), \quad & \omega_u^{c,0} &= (1-\lambda_h(\theta))(1-\omega_u), \end{aligned} \tag{50}$$

and Δ and R aggregate the four type-state credit surpluses with these weights. All cash-side equations of Section 2.6 carry over unchanged, and the model nests Ext 2 exactly at $\phi_e = \phi_u = 0$.

The hump moves in opposite directions depending on which side pays the cost (Figure 6). Symmetric and employed-only frictions amplify the hump—Δu rises from 0.288 pp at $\phi = 0$ to 0.429 pp at $\phi = 0.05$ (symmetric) and to 0.363 pp at $\phi_e = 0.02$ (employed-only), with the peak drifting slightly upward in both cases. The unemployed-only friction reverses direction: as ϕ_u rises from 0 to 1, Δu falls monotonically from 0.288 pp to 0.119 pp, and the peak shifts slightly leftward.

The reversal traces to the weight asymmetry in (50). At the calibrated values ($\omega_e = 0.044$, $\omega_u = 0.5$, $\lambda_h \approx 0.59$, $\delta \approx 0.042$), the employed credit weight in state 1 is $\omega_e^{c,1} \approx 0.92$, dwarfing the unemployed weight in the same state ($\omega_u^{c,1} \approx 0.02$); in state 0 the two weights are closer (0.56 vs. 0.21) but still asymmetric. Taxing the unemployed credit margin ($\phi_u > 0$) therefore depresses $\mathcal{S}_u^{cb}$ most, drops Ω_0 (the value of being unemployed) more than Ω_1, raises $\Delta = \Omega_1 - \Omega_0$, lowers the wage (40), and raises vacancy creation through free entry (34). The unemployed-only friction acts as a pressure-release valve on the very mechanism that drives the hump. Taxing the employed credit margin reverses every step of this calculation: Ω_1 falls, Δ drops, the wage rises, vacancy creation contracts, and the hump grows.

Crucially, ϕ_u depresses Ω_0 not just by the direct cost $\phi_u q_c^u$ but by the *quantity distortion*: credit trades shrink below $q^\star$ through the modified FOC (49), and the unrecovered surplus $T(q^\star) - T_u(q_c^u)$ is a Lagos-Wright-style deadweight that compounds with the direct cost. Without this quantity response, ϕ_u would act as a simple lump-sum tax with much weaker effect on Δ.

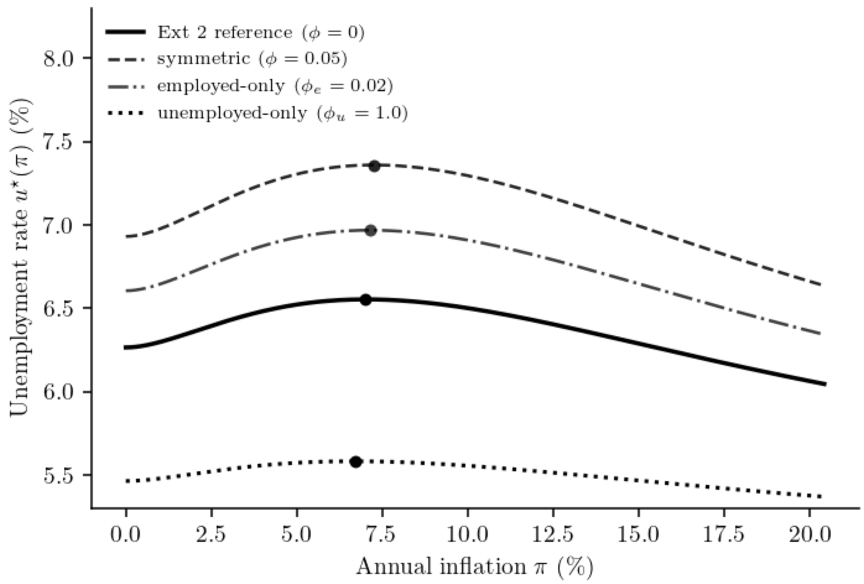


FIGURE 6. Long-run inflation–unemployment relationship under Ext 2.1 in the three focal scenarios, at $\omega_u = 0.5$ throughout. Filled dots mark interior peaks. Symmetric and employed-only frictions amplify the hump; the unemployed-only friction shrinks it.

5. Conclusion

We add differential credit access to a labour-market matching model with money: employed workers transact on credit with positive probability, while unemployed workers do not. This single asymmetry is sufficient to generate a non-trivial long-run relationship between inflation and unemployment.

The main result is that inflation operates on equilibrium unemployment through two channels of opposite sign. A Liquidity Channel depresses firm KW profit as cash quantities shrink with inflation, pushing unemployment up. An Employment Channel raises the employment premium as the cash-use asymmetry between AD-employed and AD-unemployed workers widens, lowering wages and pushing unemployment down. At low inflation the first channel dominates and unemployment rises; at high inflation the second takes over and unemployment falls. The two channels together produce a hump-shaped long-run inflation–unemployment relationship generated entirely by the credit-access mechanism, without recourse to nominal rigidities.

Three extensions sharpen the mechanism. Endogenising the cash probability—letting the share of credit users respond to inflation—pulls the hump's peak forward and dampens its amplitude, but does not eliminate it. Extending credit access to the unemployed monotonically shrinks the hump as the asymmetry between the two employment types closes, and erases it altogether at the symmetric limit where the Employment Channel collapses. Adding a per-trade credit cost reverses the macro outcome depending on which side bears it: a cost concentrated on the unemployed credit margin flattens the hump rather than amplifying it, because the unemployed-side margin enters the unemployed value disproportionately through the Lagos–Wright quantity distortion. Together the three exercises map the depth, breadth, and quality of credit infrastructure onto the geometry of the long-run inflation–unemployment relationship.

Two items shape the agenda for the next draft. First, the cross-extension welfare comparison is preliminary: it relies on a single focal calibration and does not span the joint parameter plane of credit breadth and credit quality that a full policy reading would require. Second, the cross-country mapping suggested by the employed-versus-unemployed weight asymmetry is empirically testable in principle, and identifying economies that exhibit the two-tier credit configuration is the natural empirical follow-up.

A. Proofs

A.1. Proof of Proposition 1

The proof proceeds in five steps. Throughout, the maintained assumptions on $(u, c, \lambda_h, \lambda_f)$ of Section 2.1 are imposed: $u' > 0$, $u'' < 0$, $c' \geq 0$, $c'' \geq 0$, the Inada condition $u'(0^+) = +\infty$, and $c'(0^+) \in [0, 1]$.

Step 1 (Properties of h). The Kalai payment schedule (3) satisfies $g'(q) = \eta u'(q) + (1-\eta)c'(q)$, so

$$h(q) \equiv \frac{u'(q)}{g'(q)} - 1 = \frac{(1-\eta)\left[u'(q) - c'(q)\right]}{\eta u'(q) + (1-\eta)c'(q)}. \tag{51}$$

The numerator $u'(q) - c'(q)$ is strictly decreasing in q (since u' is strictly decreasing and c' is non-decreasing), vanishes at $q^\star$ (by the definition of $q^\star$), and tends to $+\infty$ as $q \to 0^+$ (by Inada). The denominator is strictly positive on $(0, \infty)$. Therefore h is continuous and strictly decreasing on $(0, q^\star]$ with

$$h(q^\star) = 0, \qquad \lim_{q \to 0^+} h(q) = \frac{1-\eta}{\eta}.$$

The right-hand limit is finite because, as $q \to 0^+$, $g'(q) \sim \eta u'(q)$ and so $u'(q)/g'(q) \to 1/\eta$.

Step 2 (The employed money-demand equation has a unique solution). The employed money-demand equation (36) reads $i = \alpha_h w_1 h(q_1^m)$ with the cash-use weight $w_1 = \delta + (1-\delta)\omega_e \in (\omega_e, 1]$. By Step 1, the map $h : (0, q^\star] \to [0, (1-\eta)/\eta)$ is a strictly decreasing bijection onto its image. Hence (36) admits a unique $q_1^m \in (0, q^\star]$ whenever

$$i \;<\; \bar{i}_1 \;\equiv\; \alpha_h\, w_1\, \frac{1-\eta}{\eta}. \tag{52}$$

This solution is independent of θ.

Step 3 (The unemployed money-demand equation has a unique solution for every θ*).* The unemployed money-demand equation (37) reads $i = \alpha_h w_0(\theta)\, h(q_0^m)$ with $w_0(\theta) = \lambda_h(\theta)\omega_e + 1 - \lambda_h(\theta)$. Since $\lambda_h(\theta) \in [0,1]$, the weight $w_0(\theta) \in [\omega_e, 1]$ for all $\theta \in (0,\infty)$, and $w_0(\theta) \geq \omega_e$ uniformly in θ. Therefore (37) admits a unique $q_0^m(\theta; i) \in (0, q^\star]$ whenever

$$i \;<\; \bar{i} \;\equiv\; \alpha_h\, \omega_e\, \frac{1-\eta}{\eta}. \tag{53}$$

Because $w_1 \geq \omega_e$, we have $\bar{i}_1 \geq \bar{i}$, so the bound $\bar{i}$ of the proposition is the binding constraint. Continuity of w_0 in θ, together with continuity of h^{-1} on its image, implies that $q_0^m(\cdot\,; i)$ is continuous on $(0,\infty)$.

Step 4 (Existence of $\theta^\star$*).* Beveridge gives $u(\theta) = \delta/[\delta + \lambda_h(\theta)]$, continuous and strictly decreasing in θ, mapping $(0,\infty)$ onto a subset of $(0,1)$. Substituting $u(\theta)$ and $q_0^m(\theta; i)$ into Δ (25), R (32), and $S = y - b + R + \Delta$ yields continuous maps of θ. Hence

$$\theta \;\longmapsto\; \mathcal{F}(\theta; i) \;\equiv\; k - \Xi(\theta)\,\lambda_f(\theta)\,S\big(i, u(\theta), \theta\big)$$

is continuous on $(0,\infty)$. The boundary behaviour is:

- As $\theta \to 0^+$: $\lambda_h(\theta) \to 0$ and $\lambda_f(\theta) \to +\infty$ by the standard matching assumption. Hence $u(\theta) \to 1$, $w_0(\theta) \to 1$, and the limit of $q_0^m(\theta; i)$ is the unique solution of $i = \alpha_h h(q_0^m)$, which lies in $(0, q^\star]$ since $i < \bar{i} \leq \alpha_h(1-\eta)/\eta$. The factor $\Xi(\theta) \to \beta(1-\gamma)/A > 0$ and the surplus $S(i, u(\theta), \theta)$ converges to a finite limit which is positive because $y > b$, $R \geq 0$, and the credit-trade contribution to Δ remains finite and non-negative (its coefficient $(1-\delta)(1-\omega_e)$ is positive and θ-independent, and $\mathcal{S}^{cb} \geq 0$). Therefore $\Xi(\theta)\lambda_f(\theta)S \to +\infty$ and $\mathcal{F}(\theta; i) \to -\infty$.
- As $\theta \to +\infty$: $\lambda_f(\theta) \to 0$ by the standard matching assumption, $\Xi(\theta)$ remains bounded above (since $B(\theta) \to A + \beta$), and the cash quantities, surpluses, u, and hence S all remain bounded. Therefore $\Xi(\theta)\lambda_f(\theta)S \to 0$ and $\mathcal{F}(\theta; i) \to k > 0$.

By continuity and the intermediate value theorem, there exists $\theta^\star \in (0,\infty)$ with $\mathcal{F}(\theta^\star; i) = 0$, establishing part (a).

Step 5 (Uniqueness under strict monotonicity). Under hypothesis (b) of Proposition 1, $\mathcal{F}(\cdot\,; i)$ is strictly monotone on $(0,\infty)$. To see this explicitly under Cobb–Douglas matching, let

$G(\theta) \equiv \Xi(\theta)\lambda_f(\theta)$. With $\lambda_h(\theta) = Z\theta^{\sigma}$, $\lambda_f(\theta) = Z\theta^{\sigma-1}$, and $B(\theta) = A + \beta\,\lambda_h(\theta)$,

$$G(\theta) \;=\; \frac{\beta(1-\gamma)\,Z\theta^{\sigma-1}}{A+\gamma\beta Z\theta^{\sigma}}, \qquad \frac{d\ln G(\theta)}{d\theta} \;=\; -\frac{1}{\theta}\left\{(1-\sigma) + \frac{\sigma\gamma\beta\lambda_h(\theta)}{A+\gamma\beta\lambda_h(\theta)}\right\} \;<\; 0,$$

so G is strictly decreasing on $(0,\infty)$. Differentiating $\mathcal{F} = k - G(\theta)S(\theta;i)$ gives $\partial\mathcal{F}/\partial\theta = -G'(\theta)S - G(\theta)\,\partial S/\partial\theta$, and since $G > 0$ and $-G'/G$ equals the right-hand side of (43) divided by θ, $\partial\mathcal{F}/\partial\theta > 0$ whenever

$$\frac{1}{S}\frac{\partial S}{\partial\theta} \;<\; -\frac{G'(\theta)}{G(\theta)} \;=\; \frac{1-\sigma}{\theta} + \frac{\sigma\gamma\beta\lambda_h(\theta)}{\theta\,[A+\gamma\beta\lambda_h(\theta)]},$$

which is exactly the bound (43). Under hypothesis (b) this monotonicity holds at every θ, and combined with the boundary signs of Step 4 it forces the zero $\theta^{\star}$ to be unique. The bound is automatically satisfied at $\omega_e = 1$ (Corollary 1) and at $i = 0$ (Corollary 2), because in both cases the cash-use weights align, $\Delta \equiv 0$, R is θ-independent, and so $\partial S/\partial\theta = 0$. Given $\theta^{\star}$, Step 3 yields $q_0^{m\star} = q_0^{m}(\theta^{\star};i)$, Beveridge yields $u^{\star} = u(\theta^{\star})$, and (25), (32), (40) pin down $\Delta^{\star}$, $R^{\star}$, and $w^{\star}$ algebraically. The tuple $(q_1^{m\star}, q_0^{m\star}, \Delta^{\star}, R^{\star}, w^{\star}, u^{\star}, \theta^{\star})$ is therefore the unique SME at i. □

A.2. Proof of Lemma 1

From the money-demand equations (36)–(37), implicit differentiation with respect to i at fixed θ gives

$$\frac{\partial q_j^m}{\partial i} \;=\; \frac{1}{\alpha_h\, w_j\, h'(q_j^m)} \;<\; 0, \qquad j \in \{0,1\},$$

since $h'(q) < 0$ on $(0, q^{\star}]$ (Step 1 of Appendix A.1). Differentiating $R(i,u,\theta)$ from (32) at fixed (u,θ), the credit-trade term $(1-u)(1-\omega_e)\mathcal{S}^{cf}$ has zero i-derivative; the cash-trade terms yield

$$\frac{\partial R}{\partial i} \;=\; \alpha_0\left[(1-u)\,w_1\,\mathcal{S}^{mf\prime}(q_1^m)\,\frac{\partial q_1^m}{\partial i} \;+\; u\,w_0(\theta)\,\mathcal{S}^{mf\prime}(q_0^m)\,\frac{\partial q_0^m}{\partial i}\right].$$

Since $\mathcal{S}^{mf}(q) = \eta[u(q) - c(q)]$ satisfies $\mathcal{S}^{mf\prime}(q) = \eta[u'(q) - c'(q)] > 0$ on $(0, q^{\star})$ (by the definition of $q^{\star}$ and the monotonicity of u' and c'), each summand is the product of a strictly positive prefactor and a strictly negative derivative. Therefore $\partial R/\partial i < 0$. □

A.3. Proof of Lemma 2

From the money-demand equations (36)–(37), $w_1\, h(q_1^m) = w_0(\theta)\, h(q_0^m) = i/\alpha_h$. The difference between the two weights is

$$w_1 - w_0(\theta) \;=\; \delta + (1-\delta)\omega_e \;-\; \big[\lambda_h(\theta)\omega_e + 1 - \lambda_h(\theta)\big] \;=\; (1-\omega_e)\big[\lambda_h(\theta) - (1-\delta)\big].$$

Beveridge (42) gives $\lambda_h(\theta) = \delta(1-u)/u$, so $\lambda_h(\theta) - (1-\delta) = (\delta - u)/u$. Hence $w_1 < w_0(\theta)$ if and only if $u > \delta$. Under this condition, since h is strictly decreasing on $(0, q^\star]$, $w_1\, h(q_1^m) = w_0(\theta)\, h(q_0^m)$ forces $h(q_1^m) > h(q_0^m)$ and therefore $q_1^m < q_0^m$. Both quantities lie in $(0, q^\star]$ with $q_j^m = q^\star$ iff $i = 0$ (since $h(q^\star) = 0$). Finally, at $\omega_e \to 1$ the difference collapses: $w_1 = w_0 = 1$ and the two money-demand equations coincide. □

A.4. Proof of Corollary 3

At $\omega_e = 1$: $w_1 = \delta + (1-\delta)\cdot 1 = 1$ and $w_0(\theta) = \lambda_h(\theta)\cdot 1 + 1 - \lambda_h(\theta) = 1$, so the two money-demand equations (36)–(37) coincide and $q_1^m = q_0^m \equiv q^m(i)$. The credit coefficient $(1-\omega_e)$ in (44) vanishes, so $\Delta(i,\theta) = \alpha_h\big[\mathcal{S}^{mb}(q^m) - \mathcal{S}^{mb}(q^m)\big] = 0$ for every (i,θ). The joint surplus becomes $S = y - b + R$. Similarly, the credit term in R (32) vanishes, leaving $R = \alpha_0 \mathcal{S}^{mf}(q^m(i))$ with $\partial R/\partial i < 0$ strictly. Free entry $k = \Xi(\theta)\lambda_f(\theta)\,(y - b + R)$ therefore implies $u^\star(\pi)$ is monotone increasing in π, recovering the monotone inflation–unemployment relationship of Berentsen, Menzio, and Wright (2011). □

B. Calibration details

This appendix reports the calibration strategy, the data targets and their sources, the parameter values, and the moment-matching performance underlying the baseline results of Section 3.2.

We adopt a standard internal–external split. Six parameters are *fixed externally* from the labour-search and monetary-search literatures: the quarterly discount factor $\beta = 0.9928$ (4% annual real rate), the quarterly job-separation rate $\delta = 0.042$ (CPS aggregate flow rate, FRED), the matching elasticity $\sigma = 0.72$ (Berentsen, Menzio, and Wright 2011), the household and firm KW meeting probabilities $\alpha_h = \alpha_0 = 0.40$ (Berentsen, Menzio, and Wright 2011), the production-cost curvature $\gamma_c = 1$ (linear cost as in BMW), and the AD endowment $y = 1$ (price normalisation). Eight parameters are *internally calibrated* by jointly matching eight empirical targets: the KW Kalai seller share η, the utility curvature α and scale A_u, the unemployment income b, the vacancy posting cost k, the matching efficiency Z, the employed cash probability ω_e, and the worker labour-bargaining share γ. The functional forms of the matching and utility-cost primitives are the standard Berentsen,

TABLE 1. Calibration targets

Moment	Symbol	Value	Source
Goods-market markup	markup	0.20	De Loecker, Eeckhout, and Unger (2020)
Money-demand semi-elasticity	$\partial \log(M/Y)/\partial i$	−0.51	Berentsen, Menzio, and Wright (2011)
Money/output ratio (annual)	M/Y	0.232	Lucas and Nicolini (2015)
Unemployment at Friedman rule	u_{FR}	5.87%	FRED long-run mean
Replacement ratio	b/w	0.40	Hall and Milgrom (2008)
Labor-market tightness	θ_{FR}	0.95	BLS V/U
Employed cash share	emp. cash	0.08	Greene, Schuh, and Stavins (2020)
Vacancy posting cost	k	0.50	Silva and Toledo (2009)

Menzio, and Wright (2011) choices,

$$M(u,v) \;=\; Z\,u^{1-\sigma}\,v^{\sigma}, \qquad u(q) \;=\; A_u\,\frac{q^{1-\alpha}}{1-\alpha}, \qquad c(q) \;=\; q^{\gamma_c},$$

truncated at one for the matching probabilities $\lambda_h(\theta) = Z\theta^{\sigma}$ and $\lambda_f(\theta) = Z\theta^{\sigma-1}$ to remain admissible.

The eight empirical targets are listed in Table 1. The goods-market markup target follows De Loecker, Eeckhout, and Unger (2020); the money-demand semi-elasticity and the annual money/output ratio are taken from the new-monetarist calibration tradition of Berentsen, Menzio, and Wright (2011) and Lucas and Nicolini (2015); the Friedman-rule unemployment rate is the long-run U.S. CPS mean reported on FRED; the worker outside-utility-to-wage ratio follows Hall and Milgrom (2008); the equilibrium labour-market tightness is the long-run BLS vacancy-to-unemployment ratio; the employed cash-transaction share is taken from the Federal Reserve Bank of Atlanta's Diary of Consumer Payment Choice (Greene, Schuh, and Stavins 2020, SCPC/DCPC 2015–2019); the vacancy posting cost is the Silva–Toledo estimate (Silva and Toledo 2009). All targets are at quarterly model frequency unless noted; the money–output ratio and its semi-elasticity are reported annually.

Both blocks of parameters are reported in Table 2, and the model-implied moments are compared with their targets in Table 3. The model is solved at each candidate parameter vector by the bisection algorithm of Proposition 1: given i, the employed money-demand equation (36) is inverted directly for q_1^m, and a bisection over θ on the free-entry residual $\mathcal{F}(\theta; i)$ delivers $\theta^\star$, $u^\star = u(\theta^\star)$, and $q_0^{m\star}$. The eight moments are evaluated at the Friedman-rule equilibrium for the labour-market quantities and at the mean nominal rate $\bar{\imath}$ (annual inflation 5.82%) for the money-demand moments.

TABLE 2. Calibrated and fixed parameters

Parameter	Symbol	Value
Internally calibrated		
Seller bargaining share (KW Kalai)	η	0.275
Utility curvature	α	0.333
Utility scale	A_u	3.113
Unemployment income	b	1.000
Vacancy posting cost	k	0.500
Matching efficiency	Z	0.652
Employed cash probability	ω_e	0.044
Worker labour bargaining share	γ	0.760
Fixed externally		
Discount factor (quarterly)	β	0.9928
Separation rate (quarterly)	δ	0.042
Matching elasticity	σ	0.72
KW meeting prob. (household / firm)	α_h, α_0	0.40
Cost curvature	γ_c	1.00
AD endowment	y	1.00

C. Welfare cost of inflation at the baseline

We measure the welfare cost of inflation in consumption-equivalent units (CEV). Define the per-period aggregate welfare flow at an interior stationary equilibrium $(q_1^{m\star}, q_0^{m\star}, u^\star, \theta^\star)$ as the sum of net AD output, expected KW gains from trade, and minus vacancy-posting costs:

$$\begin{aligned} \mathcal{W}(\pi) \equiv\ & (1-u^\star)\,y \;-\; k\theta^\star u^\star \\ & + \alpha_h\Big[(1-u^\star)\,w_1\,\mathcal{T}(q_1^{m\star}) \;+\; u^\star\,w_0(\theta^\star)\,\mathcal{T}(q_0^{m\star}) \;+\; (1-u^\star)(1-\omega_e)\,\mathcal{T}(q^\star)\Big], \end{aligned} \tag{54}$$

where $\mathcal{T}(q) \equiv u(q) - c(q)$ is the total trade surplus per KW meeting and the cash-use weights $w_1, w_0(\theta)$ are from (29). The CEV welfare cost relative to the Friedman rule is

$$\xi(\pi) \;\equiv\; \frac{\mathcal{W}(0) - \mathcal{W}(\pi)}{\mathcal{W}(0)}, \tag{55}$$

expressed as a percentage of Friedman-rule output.

Figure 7 plots $\xi(\pi)$ at the calibrated baseline together with its BMW limit ($\omega_e \to 1$). The welfare cost is strictly increasing in π over the entire interior monetary regime, even though $u^\star(\pi)$ itself is hump-shaped: the fall in $u^\star$ above $\pi^\star$ does not deliver welfare gains because it is purchased through smaller cash-trade quantities, which dominate total

TABLE 3. Moment matching at the calibrated baseline

Moment	Model	Target
Markup	0.185	0.200
M/Y semi-elasticity	−0.499	−0.510
M/Y ratio	0.212	0.232
u_{FR}	6.27%	5.87%
b/w	0.380	0.400
θ_{FR}	0.951	0.950
Employed cash share	0.084	0.080
Vacancy cost k	0.500	0.500

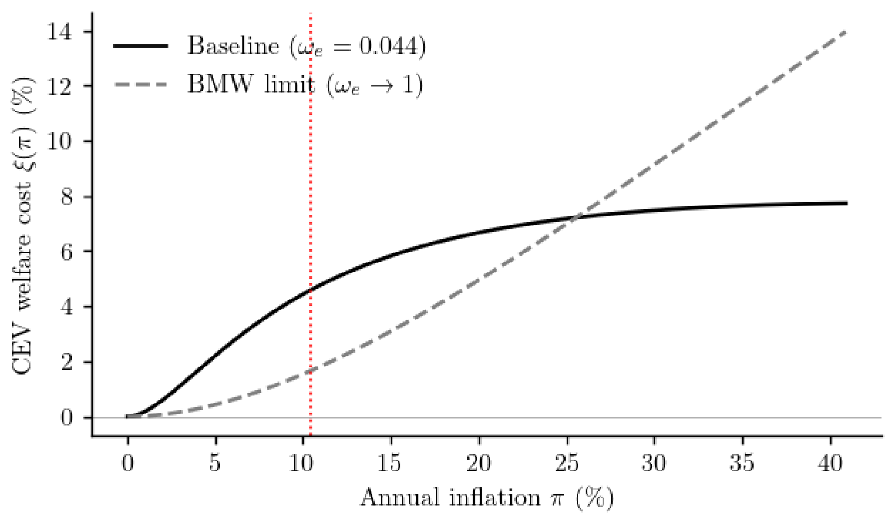


FIGURE 7. Consumption-equivalent welfare cost of inflation $\xi(\pi)$ at the calibrated baseline (solid black) and in the BMW limit $\omega_e \to 1$ (dashed gray), in percent of Friedman-rule output. Dotted vertical line: $\pi^\star = 10.42\%$.

surplus. At the baseline, $\xi(\pi)$ saturates at roughly 8% of Friedman-rule output as cash quantities $q_j^m \to 0$ and the remaining KW surplus is generated by credit trades alone; the two curves cross near $\pi \approx 25\%$, beyond which asymmetric credit access shields the baseline economy from the cash inflation tax that the BMW limit continues to bear. At policy-relevant inflation rates of 2% and 5%, $\xi(\pi)$ equals 0.59% and 2.20%, comparable in magnitude to the welfare costs reported in Lucas (2000); Lucas and Nicolini (2015) for the U.S. economy; at the unemployment-curve peak $\pi^\star = 10.42\%$ the cost is 4.62% of Friedman-rule output.

D. Welfare cost across extensions

We extend the same CEV measure to each of the three extensions and compare them in Figure 8. Each curve uses its *own* Friedman-rule welfare as the reference, so the four $\xi(\pi)$ curves all pass through zero at $\pi = 0$ and report within-extension welfare loss from inflation. The extensions are evaluated at the calibrated parameters of Section 3.2 together with their extension-specific primitives (Ext 1: $\kappa_{\max} = 0.495$, calibrated to nest the baseline at $\pi = \bar{\pi}$; Ext 2: $\omega_u = 0.5$; Ext 2.1: $\omega_u = 0.5$, $\phi_e = 0$, $\phi_u = 0.20$, the focal counter-intuitive case of Section 4.3). Each extension modifies the KW component of $\mathcal{W}(\pi)$ in a transparent way: in Ext 1 the cash probability ω_e becomes inflation-dependent and a steady-state adoption-cost outlay enters the welfare flow with a negative sign; in Ext 2 an additional unemployed credit branch is added and the cash-use weights generalise to (47); in Ext 2.1 the credit per-trade surplus $\mathcal{T}(q^\star)$ is replaced by the type-specific $T_j(q_c^j) = u(q_c^j) - c(q_c^j) - \phi_j q_c^j$, so the ϕ_j deadweight enters the welfare ledger directly.

Figure 8 shows the four CEV curves. Three patterns stand out. First, Ext 1 delivers a substantial reduction in welfare cost at every inflation rate beyond the calibration point: at $\pi = 15\%$ the CEV falls from 5.81% under the baseline to 4.09% under Ext 1, a reduction of roughly 30% even after accounting for the amortised adoption outlay. Endogenous credit adoption shields the economy from the cash-distortion tax even when no policy instrument is applied. Second, Ext 2 (partial credit access to the unemployed) raises the welfare cost slightly relative to the baseline at low inflation and lowers it slightly at high inflation: the symmetric Friedman-rule allocation is more efficient than the asymmetric baseline (so $\mathcal{W}(0)$ is larger), but the high-inflation distortion is essentially the same in level terms, so the percentage loss relative to a larger $\mathcal{W}(0)$ ends up slightly smaller. Third—and most importantly for the policy reading of Section 4.3 —Ext 2.1 with unemployed-only friction $\phi_u = 0.20$ is essentially indistinguishable from Ext 2 in welfare terms (the two curves differ by less than 0.05 percentage points at every inflation rate). The macro flattening of Section 4.3 therefore has a roughly neutral welfare consequence at this calibration: the gain from lower unemployment is offset by the deadweight $\phi_u q_c^u$ cost borne by the unemployed.

Two caveats apply. First, each extension's CEV uses its own Friedman-rule reference, so absolute welfare levels are not comparable across scenarios; the figure compares the *shape* of $\xi(\pi)$, not its level. Second, Ext 2.1 is shown only at the unemployed-only configuration; the symmetric and employed-only scenarios of Section 4.3 introduce deadweight on the larger employed-credit margin and would raise the welfare cost noticeably above the Ext 2 reference.

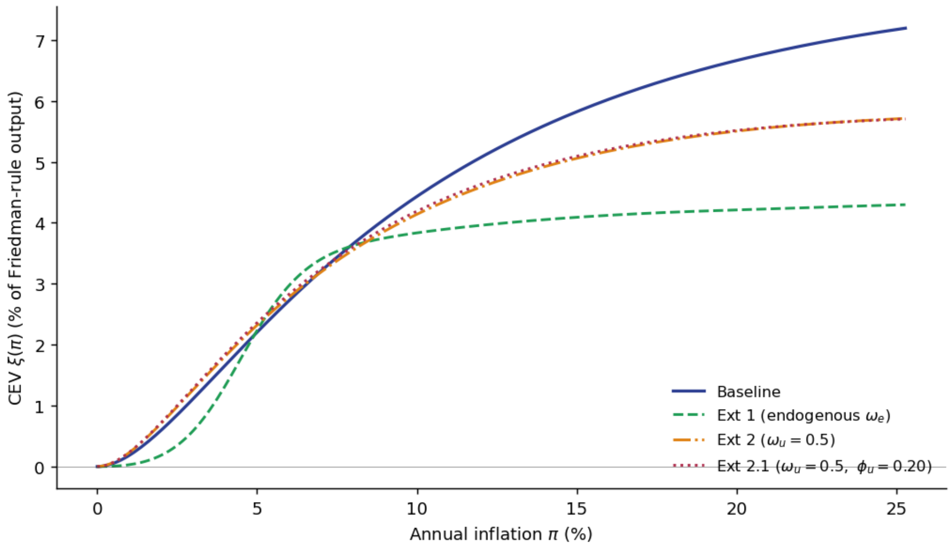


FIGURE 8. Consumption-equivalent welfare cost $\xi(\pi)$ across the baseline and the three extensions, in percent of each scenario's own Friedman-rule output. Baseline (solid blue), Ext 2 (dash-dot orange), and Ext 2.1 (dotted red) saturate near 5–5.5% above $\pi \approx 15\%$; Ext 1 (dashed green), which includes the amortised adoption-cost deduction, saturates substantially lower at roughly 4.1% because endogenous credit adoption truncates the cash-distortion channel. Ext 2 and Ext 2.1 are nearly indistinguishable in welfare terms, indicating that the macro flattening of unemployed-only ϕ_u in Section 4.3 is roughly welfare-neutral at this calibration.

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